Agriculture sector: Rice Cultivation Methane Emissions Estimated from Satellite, Statistical, and Inventory-Based Methods

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1. Introduction

The agricultural sector is a major contributor to global GHG emissions, contributing around 11.7% of total emissions; within agriculture, rice cultivation is the third-highest contributor, at over 1% of total global emissions (Ge, Friedrich and Vigna, 2024). Rice cultivation primarily emits methane (CH_4), a potent GHG with a 100-year global warming potential (GWP) 27.2 times that of carbon dioxide. Rice is generally grown in flooded paddies, creating anaerobic conditions ideal for bacteria that feed on decomposing organic matter and emit methane as a byproduct. Given that rice is a key staple crop in much of the world, it is crucial to be able to track methane production from rice cultivation and to investigate potential methods for mitigating those emissions.

Rice-cultivation methane is most commonly estimated through national inventory compilations, notably the Food and Agriculture Organization's FAOSTAT "Emissions from Crops" database and the greenhouse-gas inventories that countries report to the UNFCCC. These inventories are globally comprehensive and authoritative, but they carry important limitations for tracking rice methane. They are spatially coarse, reporting a single national total rather than resolving where within a country rice is grown; they are temporally lagged, typically trailing the current year by roughly two years and reported only at annual resolution; and they rest largely on self-reported activity data and periodic agricultural surveys rather than on direct observation and modeling of production activity. These gaps motivate an observation-based approach that resolves rice extent, cropping intensity, and seasonal timing directly.

Building on methods previously published by the authors, the Climate TRACE coalition estimates rice-cultivation methane using three complementary approaches: (1) direct, independent modeling of rice extent and emissions from 10 m Sentinel-1 and Sentinel-2 satellite observations for the major rice-producing countries; (2) a FAOSTAT harvested-area-indexed back-cast that extends the satellite record to earlier years for those countries; and (3) direct republication of FAOSTAT inventory estimates for the remaining countries that are not independently modeled. Together these approaches yield a spatially explicit, monthly, globally complete dataset for 2015–2025.

Building upon the first approach, we currently extend the methodology by developing a framework for estimating spatially explicit monthly CH_4 emissions from rice cultivation. The framework integrates global rice cropping-system mapping with temporal characterization of rice cultivation

and season-specific emission factors. This includes identifying and mapping the rice extent with multi-temporal satellite observations (Ginting et al., 2025). The mapped rice areas are then further characterized using temporal rice profiles to identify Rice Pattern Units (RPUs), cropping intensity, rice-growing seasons, and seasonal activity periods (Ginting et al., 2026). These spatial and temporal characteristics are subsequently combined with annual or season-specific CH_4 emission factors to estimate annual pixel-level emissions. Seasonal emissions are then disaggregated to the monthly scale according to the identified active cultivation months. For this temporal disaggregation, we assume that the emission factor assigned to each rice season is distributed uniformly across the active cultivation months of that season. Finally, pixel-level monthly emissions are aggregated within administrative boundaries to generate a global spatially explicit monthly and annual CH_4 emission inventory from rice cultivation.

Second, for countries not covered by the satellite, or first, approach, country-level rice cultivation data from the Food and Agriculture Organization (FAO) FAOSTAT were combined with appropriate emission factors derived from the literature. This was applied to countries that are not among the largest rice producers, where FAOSTAT national statistics provide adequate activity data; for these countries, methane is estimated from FAOSTAT harvested area combined with IPCC Tier-1 emission factors that distinguish flooded, rainfed, and upland production systems, and the estimates are reported at annual resolution.
These approaches correspond to the three categories of estimation listed for every country-year in the inventory and coded in Table S1: direct 10 m satellite observation; a FAOSTAT harvested-area-indexed back-cast for the years preceding satellite modeling, which accounts for the majority of country-years in the 2015–2025 series; and direct FAOSTAT republication for countries that are not independently modeled.

In all, the three approaches described here create a consistent and complete dataset that provides rice cultivation methane emissions for 2015 to 2025, provided at monthly resolution (annual for FAOSTAT-reported countries), for 118 countries globally. Of these, 57 countries are independently modeled from 10 m satellite observations and contribute roughly 98% of the inventory's total methane; the remaining 61 countries are drawn from the FAOSTAT inventory and contribute the balance (about 2%). With this consistent dataset, we identified ways to reduce rice emissions by assigning Emission Reduction Solutions (ERSs) and suggest strategies alternate wetting and drying (AWD) and balanced fertilizer composition **to** reduce methane emissions generated by rice cultivation. These ERS strategies are discussed further in section 2.5.

2. Materials and Methods

The Climate TRACE coalition provides rice cultivation emissions estimates using multiple methods. For the approach described here, rice cultivation emission estimates were produced using Sentinel-1A synthetic aperture radar (SAR) and Sentinel-2A/B time-series data and applied to map rice field extents, rice growth patterns, and number of plantings in rice producing regions at a 10m spatial resolution. Using regional, sub-regional, and seasonal emission factors (EFs), rice emissions estimates were derived which reflect rice growing practices at the country and sub-

national level (see Tables S2 to S5). The application of these satellite measurements to map rice fields and growth was piloted in Malaysia and is documented in the publication by Rudiyanto et al. (2019) and Fatchurrachman et al. (2022). This approach was applied to 57 countries out of 60, which is shown in Table S1. These countries represent the majority of rice producing regions globally.

The methodological framework consists of two sequential and interconnected components: (A) rice cropping system mapping and (B) methane (CH_4) emission estimation and monthly disaggregation, as shown in Figure 1:

- In Part A, multi-temporal satellite observations are used to first determine where rice is cultivated, followed by characterization of the temporal rice signal to identify rice cropping patterns and Rice Pattern Units (RPUs). The number of distinct rice-growing cycles is then used to determine cropping intensity, while the temporal phenological profile is used to identify when each rice season is active and its corresponding growth stages and duration. The resulting spatial and temporal rice information provides the basis for the subsequent emission calculation.
- In Part B, annual or season-specific (where available) CH_4 emission factors (EFs) are assigned to individual rice seasons to estimate annual pixel-level emissions. The seasonal emissions are then temporally disaggregated to the monthly scale according to the identified active months of each rice season, assuming that the corresponding EF is distributed equally across its active cultivation months. Finally, pixel-level monthly emissions are summed within administrative boundaries to produce monthly and annual CH_4 emission inventories.

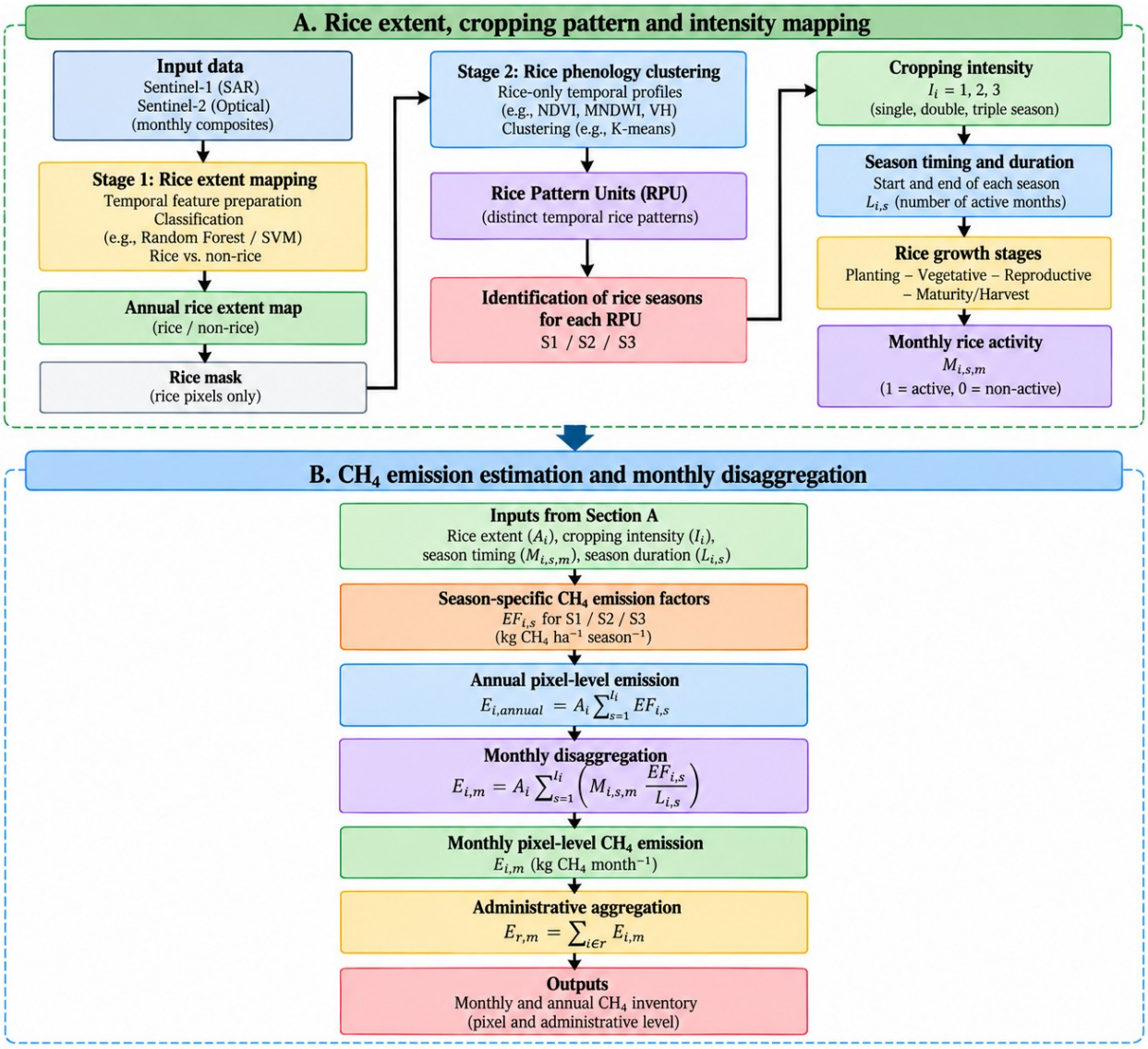


Figure 1. Overall methodological framework for spatial and monthly estimation of rice CH_4 emissions.

The workflow proceeds from multi-temporal Sentinel-1 and Sentinel-2 inputs through rice-extent masking, phenology-based clustering into Rice Pattern Units, and expert interpretation of cropping intensity and seasonal calendars (Part A), to assignment of season-specific emission factors, pixel-level annual emission estimation, monthly disaggregation, and aggregation to administrative units (Part B).

Each of the 57 countries that received the approach described above then received a ‘back-cast’ (BC) which is a FAOSTAT harvested-area-indexed back-cast, anchored to each country's first 10 m-modeled year, used for the years preceding 10 m modeling. See Section 2.4 for more information.

Lastly, for the remaining 61 countries that did not receive the above approach, FAOSTAT rice data was used. These countries cumulatively make up about 2% of global rice emissions.

2.1. Dataset

2.1.1 Satellite Data

Multi-temporal Sentinel-1 and Sentinel-2 satellite observations were used as the primary data sources for global rice cropping-system characterization. Sentinel-1 C-band synthetic aperture radar (SAR) observations provide information on rice canopy structure, surface moisture, and flooding conditions, while Sentinel-2 multispectral observations provide complementary information on vegetation dynamics and surface-water conditions. Temporal composites were generated to characterize the seasonal dynamics of rice cultivation and to reduce the effects of short-term variability and missing observations. From Sentinel-2 data, the Normalized Difference Vegetation Index (NDVI) and Modified Normalized Difference Water Index (MNDWI) were derived, while Sentinel-1 VH backscatter was used to characterize the temporal SAR response of rice fields. These temporal variables were combined to generate multi-temporal profiles used for rice extent mapping and subsequent phenology-based characterization of rice cropping patterns, intensity, and seasonal calendars. See Table 1 for additional information on uses.

2.1.2 Rice cultivation emission factors

Rice emission factors were compiled from the published literature for each modeled country, prioritizing country- or region-specific field measurements of seasonally integrated methane flux and falling back to a neighboring country's value or the IPCC default where no local study was available. Country-level emission factors and their sources are listed in Table S2, with subnational factors for China in Table S3. The Thailand and Vietnam emission factors, which are resolved at the subnational and seasonal level, are provided in Tables S4 and S5 respectively. See Table 1 for additional information on uses.

2.1.3 Ancillary Data

Ancillary datasets were used to support rice mapping, spatial aggregation, validation, and CH_4 emission estimation. Administrative boundary data were used to aggregate pixel-level emissions into administrative regions and generate regional and national inventories. Season-specific CH_4 emission factors were incorporated into the emission calculation to estimate emissions for individual rice-growing seasons. These ancillary datasets provide complementary spatial and emission information required to transform the satellite-derived rice cropping characteristics into a spatially and temporally explicit CH_4 emission inventory. See Table 1 for additional information on uses.

Table 1 Summary of datasets used, time period, and purpose.

Dataset	Resolution	Period	Purpose

Sentinel-1	10 m	1 Jan-31 Dec 2025	Rice phenology
Sentinel-2	10 m	1 Jan-31 Dec 2025	Rice phenology
Administrative boundary	Vector	—	Regional aggregation
Seasonal CH_4 EF	—	—	Emission estimation

2.2. Mapping Rice Extent, Cropping Pattern and Intensity

High-resolution mapping of rice cropping patterns, calendars, and intensities follows a sequential phenology-based framework integrating multi-temporal Sentinel-1 and Sentinel-2 observations. The approach exploits complementary temporal information from Sentinel-1 VH backscatter and Sentinel-2-derived NDVI and MNDWI to characterize the distinctive phenological trajectories of rice cultivation. The methodology first establishes the spatial extent of rice cultivation and subsequently analyzes the temporal characteristics of the mapped rice areas to identify cropping patterns, intensity, and seasonal calendars. This phenology-based framework enables the methodology to move beyond static rice extent mapping toward a dynamic characterization of rice cultivation systems.

Stage 1: Rice Extent Mapping

Multi-temporal Sentinel-1 and Sentinel-2 observations are processed to derive temporal features representing the spectral and SAR characteristics of rice cultivation. These temporal trajectories are used to distinguish rice from non-rice areas and generate a spatially continuous rice extent map. The resulting classification is converted into a binary rice mask, with rice pixels assigned a value of 1 and non-rice pixels assigned a value of 0. The rice mask is subsequently applied to the multi-temporal dataset so that only pixels identified as rice are retained for the subsequent phenological analysis. This sequential masking step reduces interference from non-rice land-cover types and ensures that the subsequent clustering focuses on variation within rice cultivation systems.

Stage 2: Rice Phenology Clustering

Following the rice extent mapping, a phenology-expert-based approach is applied to characterize the temporal dynamics of rice cultivation within the mapped rice areas. Only pixels identified as rice in Stage 1 are retained for subsequent analysis. For each rice pixel, multi-temporal Sentinel-1 and Sentinel-2 profiles are analyzed to characterize the distinctive phenological trajectories of rice cultivation. The temporal information from vegetation and water-related indicators, including NDVI, MNDWI, and VH backscatter, is used to identify characteristic rice growth cycles and distinguish different temporal cropping patterns. Similar phenological trajectories are grouped into Rice Pattern Units (RPUs) through phenology-based clustering. Each RPU represents a characteristic temporal pattern of rice cultivation and is subsequently interpreted by a phenology expert to identify the number of rice-growing seasons, cropping intensity, season timing, and growth-stage periods. The identified phenological information is then used to determine the monthly rice activity for each season, which provides the temporal basis for subsequent monthly CH_4 emission disaggregation.

Rice Pattern Unit

A Rice Pattern Unit (RPU) represents a characteristic temporal pattern of rice cultivation identified from multi-temporal satellite observations. After rice pixels are identified from the rice extent map, their temporal profiles are analyzed to characterize the phenological dynamics of rice cultivation. Pixels exhibiting similar temporal phenological characteristics are grouped into the same RPU, providing a spatially consistent representation of rice cropping patterns. The temporal profile of each RPU is then interpreted using phenology-expert knowledge to identify individual rice cultivation cycles, including the start of cultivation, active cultivation period, crop growth stages, maturity or harvest, and the onset of a subsequent cultivation cycle. Based on the number of distinct and phenologically identifiable rice cultivation cycles within a calendar year, RPUs are classified into single-season, double-season, or triple-season rice, as illustrated in Figure 2. This RPU-based characterization provides the basis for determining cropping intensity, seasonal calendars, season duration, and monthly rice activity for subsequent CH_4 emission estimation.

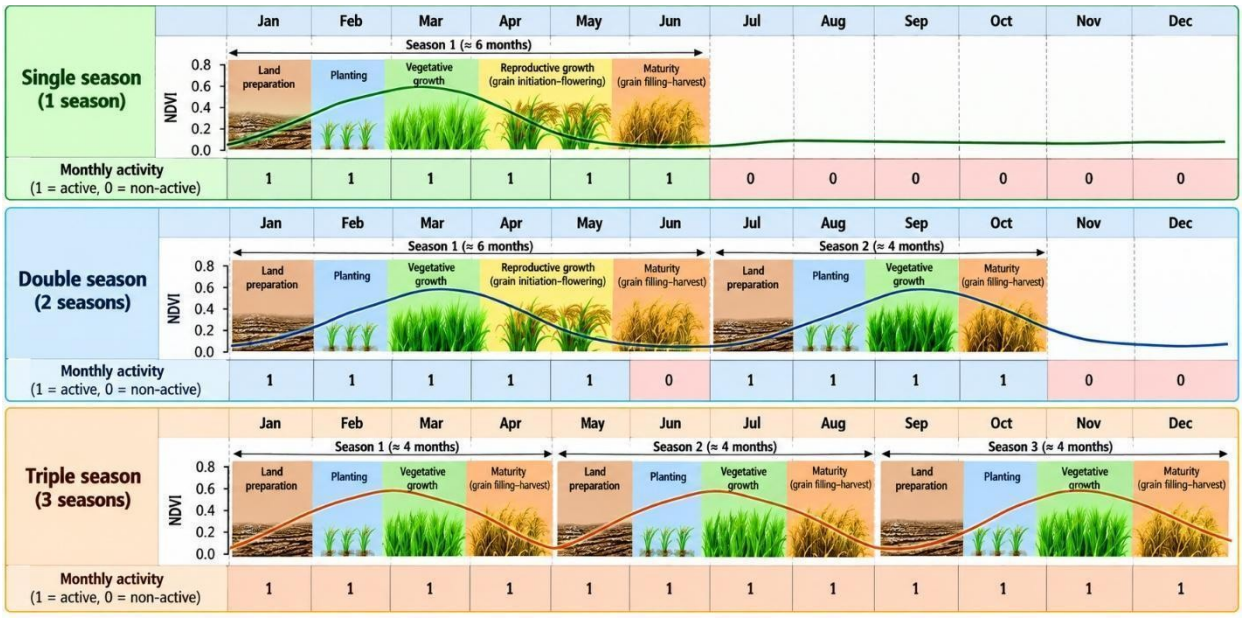


Figure 2. Phenology and cropping-pattern schematic for Rice Pattern Units (RPUs). Illustrative NDVI temporal profiles for single-, double-, and triple-season rice systems, showing rice growth stages, season duration, and monthly rice activity. Monthly activity is coded as 1 for active cultivation and 0 for non-active periods. The identified seasonal patterns provide the basis for determining cropping intensity, seasonal calendars, and monthly CH_4 emission allocation.

Cropping Intensity and Season Identification

Cropping intensity and rice-season identification are derived from the temporal phenological profiles of the Rice Pattern Units (RPUs) using a phenology-expert-based interpretation approach. For each RPU, the monthly temporal profile is visualized to identify recurrent rice-growing cycles based on characteristic changes in the spectral and SAR signals associated with rice development. The temporal profiles are interpreted according to the expected sequence of rice phenological

stages, including land preparation, planting, vegetative growth, reproductive development, maturity, and harvest. The occurrence of a complete sequence of these phenological stages is considered evidence of an individual rice-growing cycle.

As illustrated in Figure 2, an RPU may exhibit one, two, or three distinct phenological cycles within a calendar year. A single pronounced cycle is interpreted as single-season rice, two distinct cycles as double-season rice**,** and three distinct cycles as triple-season rice**.** The identification of separate cycles is based on the temporal separation between successive cultivation periods, including the decline in the crop signal following harvest and the subsequent recovery associated with the establishment of a new crop. Thus, cropping intensity is determined from the number of phenologically identifiable rice-growing cycles within the annual temporal profile:

$$I_i = N_{season,i} \quad (1)$$

where i = pixel index, Ii is the annual cropping intensity and Nseason,i is the number of identifiable rice-growing cycles within the year.

Thus:

$$I_i = \{1, \ single\text{-}season\,rice \quad 2, \ double\text{-}season\,rice \ \ 3, \ triple\text{-}season\,rice \quad (2)$$

For each identified season s, the start and end of the active cultivation period are determined from the temporal phenology profile.

Rice Growth Stages and Identification of Monthly Rice Activity and Season Duration

The temporal profile of each Rice Pattern Unit (RPU) is interpreted using the characteristic sequence of rice growth stages, including land preparation, planting, vegetative growth, reproductive growth, and maturity or harvest. These phenological stages provide the basis for identifying the beginning and end of each rice-growing season and distinguishing successive cultivation cycles within the annual temporal profile. As illustrated in Figure 2, the temporal profiles also provide the corresponding monthly representation of rice activity, where a value of 1 indicates an active rice cultivation month and 0 indicates a non-active month. The resulting monthly activity sequence explicitly represents the timing and duration of each rice season and provides the temporal basis for subsequent CH_4 emission disaggregation.

For each identified rice season s, the active cultivation months are determined from the temporal growth-stage information:

$$M_{i,s,m} = \{1, \ if\ season\ s\ is\ active \in month\ m \ \ 0, \ otherwise \quad (3)$$

This season-specific indicator preserves the identity of each rice season:

- i = pixel index

- s = season
- m = month index, m=1,…,12
- Mi,1,m = Season 1 activity
- Mi,2,m = Season 2 activity
- Mi,3,m = Season 3 activity

The duration of each rice-growing season was calculated as:

$$L_{i,s} = \sum_{m=1}^{12} \square M_{i,s,m} \qquad (4)$$

where Li,s represents the duration, in months, of rice season (s) for pixel (i).Thus, the temporal characterization provides three complementary pieces of information**:** the number of rice seasons (cropping intensity), the months in which each season is active, and the duration of each season**.** These parameters are subsequently used to allocate the season-specific CH_4 emission factor across the active months of each rice-growing season.

2.3. CH_4 Emission Estimation and Monthly Disaggregation

Rice CH_4 emissions were estimated at the pixel level by integrating the mapped rice area, annual cropping intensity, season-specific emission factors, and the identified timing and duration of rice cultivation. Each rice pixel may contain up to three distinct rice-growing seasons within a calendar year. The annual emission calculation therefore accounts for the contribution of each identifiable rice season separately, allowing the emission factor to vary among seasons. The resulting annual pixel-level emissions are subsequently disaggregated to the monthly scale using the season-specific monthly activity indicators derived from the rice phenological profiles.

Seasonal Emission Factors

A season-specific emission factor (EFi,s) is assigned to each rice-growing season (s) for pixel (i). The emission factors used in the analysis are provided in Tables S2–S5. Here, (s) denotes the rice-season index, with a maximum of three seasons per pixel ((s=1,2,3)). The use of season-specific emission factors allows the annual emission estimate to account for differences among individual rice-growing cycles.

Annual Pixel-Level CH_4 Emission

For each rice pixel (i), annual CH_4 emissions are calculated as the sum of emissions from all identifiable rice-growing seasons:

$$E_{i,annual} = A_i \sum_{s=1}^{I_i} \square EF_{i,s} \qquad (5)$$

where:

- Ei,annual = annual CH_4 emission from pixel i
- Ai = area of pixel i
- Ii = annual rice cropping intensity of pixel i, representing the number of rice-growing seasons per year
- EFi,s = CH_4 emission factor for season s
- i = pixel index
- s = rice-season index, s=1,2,3

Thus, a single-season pixel contributes one seasonal emission factor, a double-season pixel contributes two, and a triple-season pixel contributes three. The formulation explicitly accounts for differences in emission factors among individual rice-growing seasons rather than applying a single annual emission factor to all cultivation cycles.

Monthly Disaggregation

The annual seasonal emissions are subsequently disaggregated to monthly values using the season-specific monthly activity indicator (Mi,s,m). For each active month, the seasonal emission factor is assumed to be distributed equally across the active cultivation months of that season. The monthly CH_4 emission for pixel (i) is therefore calculated as:

$$E_{i,m} = A_i \sum_{s=1}^{I_i} \left(M_{i,s,m} \frac{EF_{i,s}}{L_{i,s}} \right) \quad (6)$$

where:

- Ei,m = CH_4 emission from pixel i in month m
- Mi,s,m = indicator of whether season s is active in month m
- EFi,s/Li,s = monthly allocation of the seasonal emission factor

This formulation ensures that emissions are assigned only to months in which the corresponding rice season is active.

Under this assumption, the seasonal emission factor is allocated only to months identified as active for the corresponding rice season. Months classified as non-active receive no emission contribution from that season. For pixels containing multiple rice seasons, emissions from all active seasons are summed for each month. Consequently, the monthly emission profile reflects both cropping intensity and the temporal overlap or separation of individual rice-growing seasons.

The monthly formulation is consistent with the annual calculation because summing monthly emissions over the 12 months recovers the annual pixel-level emission, Eq (5). This ensures that the temporal disaggregation redistributes the annual seasonal emissions across months without changing the annual emission total.

Aggregation to administrative regions

Following the pixel-level monthly estimation, emissions are aggregated within each administrative region. For administrative region (r), monthly CH_4 emissions are calculated as:

$$E_{r,m} = \sum_{i \in r} A_i \sum_{s=1}^{I_i} \left(M_{i,s,m} \frac{EF_{i,s}}{L_{i,s}} \right) \qquad (7)$$

Where Er,m is the total CH_4 emission in administrative region (r) during month (m), and the summation includes all rice pixels located within the region. Annual administrative emissions are obtained by summing the monthly emissions:

$$E_{r,annual} = \sum_{m=1}^{12} E_{r,m} \qquad (8)$$

This procedure produces a spatially explicit inventory of monthly and annual rice CH_4 emissions, from the individual pixel level to administrative regions and ultimately to the global scale.

2.4. Historical Reconstruction, Backfill, and Temporal Allocation

The satellite-based framework above produces native monthly estimates for the countries and years modeled at 10 m resolution (Table S1). Two additional processing steps extend the inventory to a complete 2015–2025 country-level series: reconstruction of earlier annual estimates, and, for years preceding native monthly modeling, allocation of annual totals across months. These steps address different gaps and carry different interpretive limits, and are described here so that users can distinguish satellite-derived estimates from statistical reconstructions and temporal allocations.

Historical reconstruction (2015–2020)

For the 27 countries with a continuous modeled series, annual emissions for 2015–2020 are reconstructed by indexing each country's modeled 2021 baseline to its national rice harvested-area trend from the Food and Agriculture Organization of the United Nations (FAO), so that the reconstructed series inherits the FAO year-to-year signal. Countries first modeled after 2021 are reconstructed over their pre-model years using the same FAO interannual change, anchored to each country's first modeled year. Countries not modeled by the Sentinel approach in any year are reported directly from the FAO rice-cultivation emissions inventory (release dated 28 October 2025) at annual resolution. These historical values are estimates reconstructed from agricultural activity data rather than independent satellite observations of rice cultivation or methane in each historical year. A national change in harvested area does not, by itself, establish that cultivation,

water management, or methane emitted per unit of activity changed by the same amount within each reporting unit, and the reconstructed series should be interpreted within these limits.

Monthly allocation (2021–2024)
For reporting assets with a native 2025 monthly profile, each annual methane total for 2021–2024 is allocated across months using that asset's 2025 monthly distribution: the reference share for a month is that month's 2025 methane estimate divided by the 2025 annual total, and the historical annual total is multiplied by each monthly share. The allocation preserves the annual total and does not recover year-specific planting dates, flooding periods, or harvest timing for the earlier years; annual trends come from the annual emissions series, while within-year timing comes from the 2025 reference profile. Where an asset lacks a direct 2025 profile, the profile of a broader administrative unit in the same country is used. This temporal allocation is separate from the native monthly satellite processing described in Sections 1.3–1.4.

Non-modeled countries and the water-regime basis
The FAO rice-cultivation inventory used for non-modeled countries applies IPCC Tier-1 emission factors with water-regime scaling. This supersedes an earlier Climate TRACE approach that applied a single flooded-rice emission factor to all FAOSTAT harvested area without distinguishing upland or rainfed systems, which overstated emissions in countries where rice is predominantly upland or rainfed. Recent modelling and synthesis studies (for example, Chen et al., 2023; Nikolaisen et al., 2023) suggest that Tier-1 inventories may, in turn, under-estimate African rice emissions; reconciling these remains an open item for a future release.

Together, these steps yield a single 2015–2025 inventory in which each country-year is tagged by estimation method (Table S1): native 10 m monthly estimates for satellite-modeled country-years, a FAOSTAT harvested-area-indexed back-cast for earlier years of modeled countries, and direct FAOSTAT values for countries that are not independently modeled. Modeled and back-cast countries are reported at monthly resolution and non-modeled countries at annual resolution. The following section summarizes the 2025 results and the full monthly series.

2.5 Incorporating ERS into model

The ERS strategy to apply to each asset was determined by satellite data supplemented with literature review. For countries in Southeast Asia, rice fields (assets) were identified as rainfed rice, irrigated rice, and/or deep-water rice. For countries outside of Southeast Asia, literature references provided the dominant type of rice production within each country. For assets and countries with >50% of rice production area under irrigation, the alternative wetting and drying (AWD) strategy was applied, and for all other assets, the Balanced Fertilizer Composition strategy was applied.

2.5.1 Alternate wetting and drying (AWD)

AWD was incorporated into the ERS dataset for rice cultivation after the emissions modeling process as an adjustment to the CH_4 emissions factor for each asset. The updated emissions factor

is applied to some percentage of the harvested area (activity) within each grid cell, depending on whether that location is known to be partially or entirely irrigated.

The AWD strategy is estimated to reduce the combined GWP-adjusted emissions of N_2O + CH_4 by an average of 46.9% compared to continuous flooding, according to a 2024 meta-analysis of studies (this accounts for a reduction in CH_4 emissions and an increase in N_2O emissions) (Zhao et al., 2024). In order to ensure both impacts were represented despite having no quantified N_2O emissions in this subsector, we reduced the CH_4 emissions of each asset with the AWD strategy by 46.9%, resulting in a new-to-old ratio of CH_4 emissions of 0.531. To account for assets with different percentages of irrigated rice, we used the following structure:

- Assets where the rice production area was observed or estimated to have been irrigated >95% were assigned the *awd95pct* strategy. The methane emissions were reduced by 46.9%, and this reduction was applied to 95% of the harvested area in these assets.
- Assets where the rice production area was observed or estimated to have been irrigated >=50% but less than 95% were assigned the *awd50pct* strategy. The emissions were reduced by 46.9%, and this reduction was applied to 50% of the harvested area in these assets.
- Assets where the rice production areas were observed or estimated to have been irrigated <50% were assigned Balanced Fertilizer Composition ERSs, as explained in the next section.

2.5.2 Balanced fertilizer composition

Balanced fertilizer composition was also incorporated after the emissions modeling process as an adjustment to the CH_4 emissions generated by the rice cultivation activity for each asset. This strategy is estimated by a 36-year field trial study to reduce CH_4 emissions by an average of 21.2% for early-season harvests and 28.6% for late-season harvests (Zhu et al., 2022). We have data on how many yearly harvests each rice production asset undergoes (either one, two, or three). Thus, we scaled these emissions reductions as follows:

- Rice fields with a single yearly harvest (early-season) were assigned the *rfrearly* strategy, with CH_4 emissions reduced by 21.2%. These reductions were assigned to 100% of the rice harvest area in the asset.
- Rice fields with two yearly harvests (early-season and late-season) were assigned the *rfrmiddle* strategy, with CH_4 emissions reduced by 24.9%, which is the average of early-season (21.2%) and late-season (28.6%) emissions reductions. These reductions were assigned to 100% of the rice harvest area in the asset.
- Rice fields with three yearly harvests (early-season, middle-season, and late-season) were assigned the *rfrlate* strategy, with CH_4 emissions reduced by 26.1%. This emissions reduction was calculated by averaging the emissions reductions for one early season (21.2%) harvest and two late season harvests (2*28.6%). These reductions were assigned to 100% of the rice harvest area in the asset.

2.5.3 Calculating emissions after incorporating ERS

Rice cultivation emissions adjusted for ERSs are calculated as shown in Equations 2 and 3.

$$EF_{ERS} = EF * R_{EF} \quad (9)$$

where EF_{ERS} is the methane emissions factor after implementing ERS,
EF is the methane emissions factor before implementing ERS,
and R_{EF} is the new-to-old emissions factor ratio.

$$emissions_{ERS} = EF_{ERS} * [A_{max} * A] + EF * [(1 - A_{max}) * A] \quad (10)$$

Where $emissions_{ERS}$ is the adjusted annual methane emissions after implementing ERS,
A_{max} is the ratio of maximum activity affected by the ERS to total activity,
and A is the annual activity of the asset.

2.5.4 Verifying ERS estimates

The proposed rice cultivation ERSs are considered scenario-based: approaches that are shown (in limited deployment and studies) to reduce methane emissions, but are not fully implemented. In the future, methods to observe and verify AWD may be implemented using remote sensing tools. Time series wetness indices combined with in-situ moisture readings have been demonstrated to provide a prediction of AWD adoption (Lovell 2019). As these practices begin deployment within carbon offset projects, it is likely that several scalable remote sensing tools may be available to cost effectively monitor implementation, but given low current adoption rates of AWD, attempts to implement and validate remote sensing detection strategies in new geographies is challenging.

The composition of fertilizer applied to rice fields cannot be directly observed, though there may be opportunities to use independent evidence via fertilizer consumption data at a country or site specific scale that could provide insight into the adoption of fertilizer-based strategies.

3. Results and Discussion

3.1. Monthly Rice Activity

Figure 3 shows the monthly active rice area derived from the identified rice-growing seasons. The results reveal distinct seasonal patterns among major rice-producing regions. Here, the monthly active rice area does not represent the total rice-growing area or harvested area. Instead, monthly active rice area is defined as the area of rice fields that is actively under rice cultivation during a particular month, based on the identified season-specific phenological activity. Thus, a rice field contributes to the active rice area only during the months assigned as active for its corresponding rice-growing season.

In the subtropics of the Northern Hemisphere**,** including North India, Northeast China, Japan, South and North Korea, the United States, and Italy, single-season rice systems generally concentrate cultivation during the warmer months. Consequently, active rice areas increase markedly during the summer growing season and decline toward winter. In contrast, the seasonal pattern is reversed in the Southern Hemisphere**,** where the principal single-season cultivation period generally coincides with the austral summer. This pattern is particularly evident in countries such as Brazil, where active rice areas are concentrated during the latter part of the calendar year and extend into the early months of the following year.

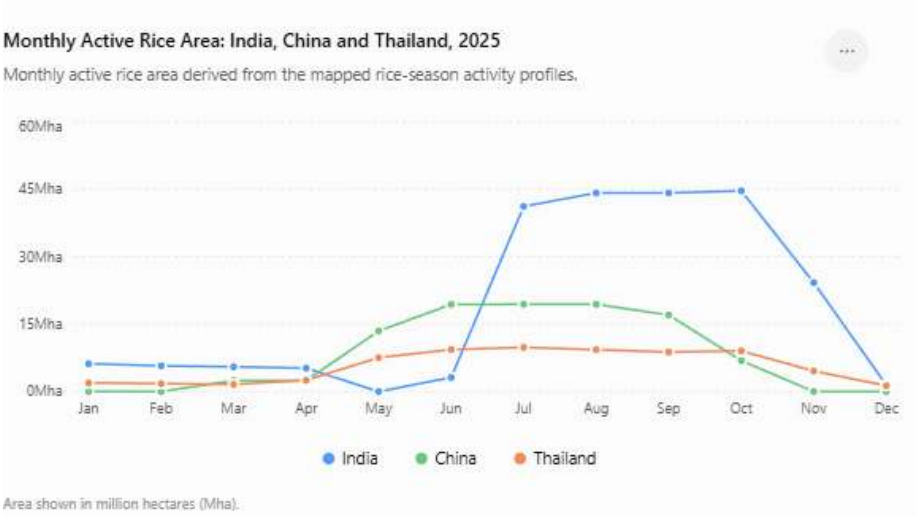
In contrast, South and Southeast Asian rice-growing systems**,** particularly those represented by India, Bangladesh, Indonesia, Vietnam, and Thailand, exhibit more complex temporal patterns because rice cultivation can occur in multiple seasons within the same year. Single-, double-, and triple-season systems produce overlapping cultivation periods, resulting in rice activity that can extend through much or nearly all of the year. Consequently, these regions do not exhibit a simple summer–winter cycle; instead, the monthly active area reflects the combined phenological activity of multiple rice-growing seasons.

The patterns observed for Tanzania and Madagascar may be more strongly associated with regional rainfall seasonality, where rice cultivation is influenced by the timing of rainy seasons. Their monthly active-area profiles therefore show stronger seasonal concentration than areas where irrigation and multiple cropping enable rice cultivation throughout much of the year. Overall, the monthly active-area profiles demonstrate the importance of explicitly representing seasonal calendars and cropping intensity rather than assuming a single global rice-growing season.

(a)



(b)



(c)

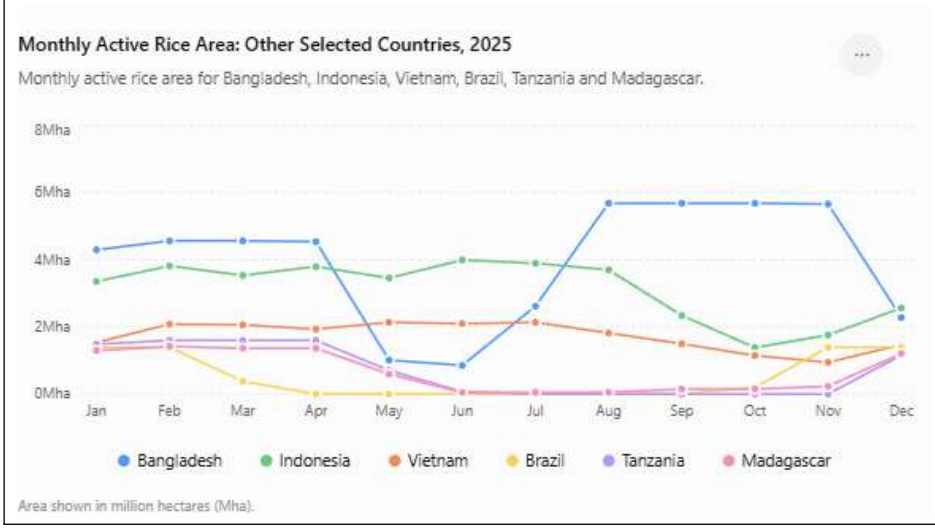


Figure 3 continued

Figure 3. Monthly active rice area derived from the identified rice-growing seasons. (a) Global active rice area; (b) India, China, and Thailand; and (c) Bangladesh, Indonesia, Vietnam, Brazil, Tanzania, and Madagascar. The monthly profiles illustrate differences in rice-growing seasonality, cropping intensity, and hemispheric and rainfall-driven cultivation patterns. Complete monthly active rice-area data for all 58 countries with monthly estimates are provided in Table S6; Hungary and Malawi are reported at annual resolution only.

3.2. Monthly CH_4 emission

3.2.1 Monthly CH_4 emission patterns

Figure 4 shows substantial temporal variation in rice-cultivation CH_4 emissions across the year. At the global scale, monthly emissions increased from 1.20 Mt CH_4 in January to 3.51 Mt CH_4 in August, followed by a gradual decline to 0.86 Mt CH_4 in December. This seasonal pattern broadly corresponds to the monthly active rice-area dynamics, which increased from 27.7 Mha in January to 108.2 Mha in August and subsequently declined toward December. The close correspondence indicates that the monthly inventory successfully transfers the temporal characteristics of rice cultivation into the emission estimates.

The relationship is not expected to be strictly proportional because the monthly emissions also depend on the season-specific emission factor (EFi,s) and the duration of each rice-growing season. Thus, two months with similar active rice areas can produce different CH_4 emissions when they contain rice seasons with different emission factors. The monthly estimates therefore represent the combined effect of where rice is actively cultivated, how many seasons are active, and the emission characteristics assigned to each season.

For India, China, and Thailand (Figure 4b), the results demonstrate contrasting seasonal patterns. India shows relatively low emissions during the first half of the year, followed by a sharp increase from July, reaching approximately 0.808 Mt CH_4 in October. This pattern is consistent with the strong increase in active rice area during the corresponding period. China shows a pronounced increase from May to August, with emissions reaching approximately 0.660 Mt CH_4 in July and August, before declining substantially toward October. Thailand maintains substantial emissions over a longer part of the year, with emissions increasing from May and remaining high through October. These patterns reflect differences in the timing and number of rice-growing seasons among the three countries.

The other countries shown in Figure 4c demonstrate additional examples of seasonal and multi-season rice systems. Bangladesh and Indonesia maintain substantial CH_4 emissions over much of the year, reflecting the presence of multiple rice-growing seasons and relatively prolonged rice activity. Indonesia, for example, maintains emissions above approximately 0.269 Mt CH_4 during January–August, with a subsequent decline before increasing again in December. Vietnam also exhibits relatively persistent emissions throughout the year, although the magnitude declines

during September–November. These patterns are consistent with the broader temporal distribution of active rice cultivation in South and Southeast Asia.

In contrast, Brazil exhibits a more strongly seasonal pattern, with emissions concentrated during the beginning and end of the calendar year and little or no estimated emission during April–September. This reflects the seasonal nature of its dominant rice cultivation cycle and the Southern Hemisphere growing season. Tanzania and Madagascar similarly show stronger seasonal fluctuations, with their emission profiles consistent with cultivation periods associated with seasonal rainfall and local cropping calendars.

Overall, Figure 4 demonstrates that the proposed framework converts spatially mapped rice cropping patterns into a temporally explicit CH_4 emission inventory. Rather than distributing annual emissions uniformly across the 12 months, emissions are assigned to months in which rice cultivation is identified as active for each season. This preserves the spatial and seasonal heterogeneity of rice production and provides a more informative representation of the timing of rice-cultivation CH_4 emissions.

(a)

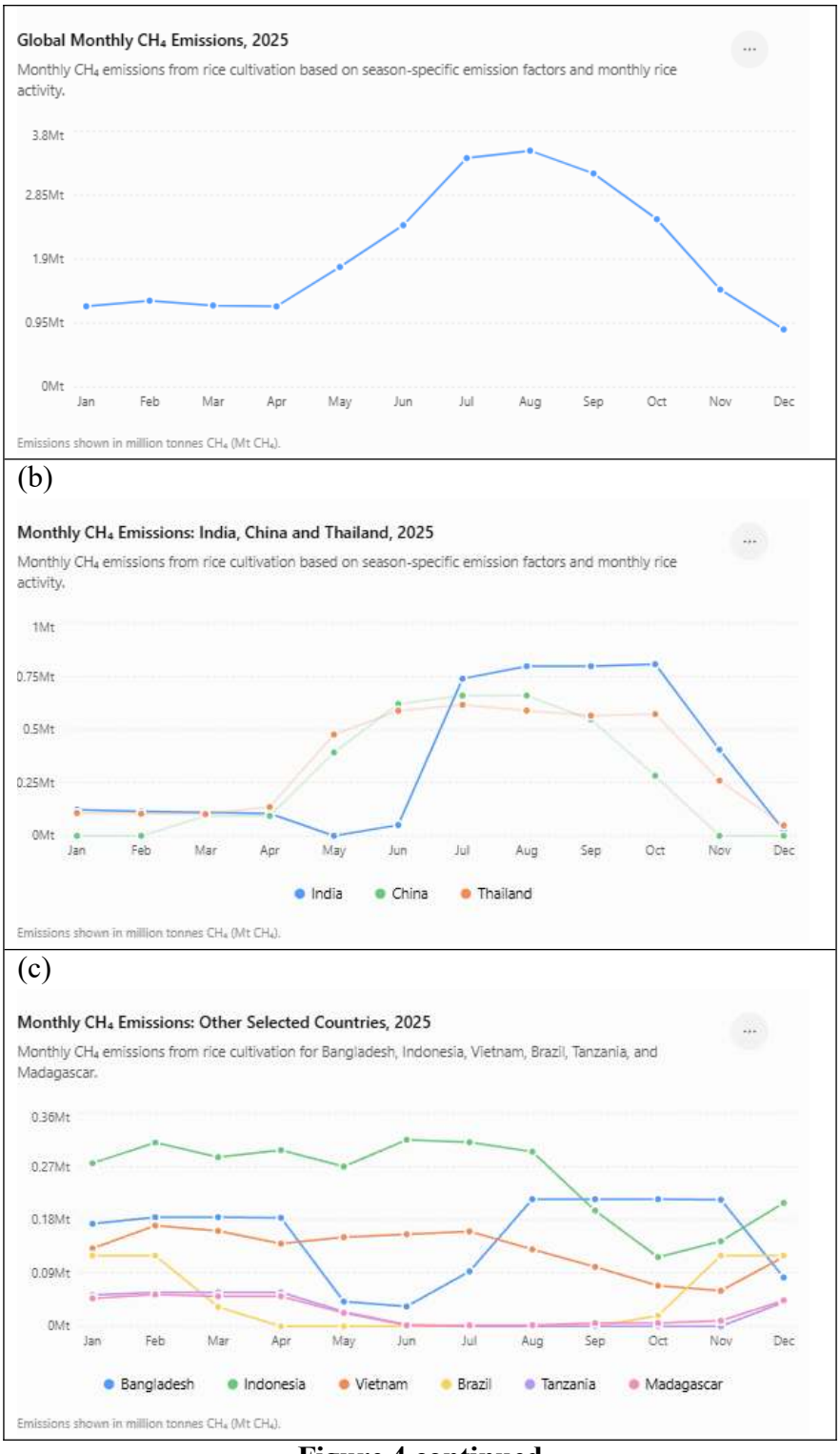


Figure 4 continued

Figure 4. Monthly CH_4 emissions from rice cultivation in 2025. (a) Global monthly CH_4 emissions; (b) India, China, and Thailand; and (c) Bangladesh, Indonesia, Vietnam, Brazil, Tanzania, and Madagascar. The monthly emission profiles reflect the temporal distribution of active rice cultivation combined with season-specific CH_4 emission factors. Complete monthly CH_4 emission data for all 58 countries with monthly estimates are provided in Table S7; Hungary and Malawi are reported at annual resolution only.

3.2.2 Annual and Monthly CH_4 Emissions in 2025

The estimated global rice-cultivation CH_4 emission for 2025 was 23.97 million tonnes (Mt CH_4). The annual total was distributed unevenly throughout the year, with monthly emissions ranging from 0.86 Mt CH_4 in December to 3.51 Mt CH_4 in August. Emissions increased substantially from May onward, reaching the annual maximum in August, before declining toward the end of the year. This temporal pattern broadly follows the distribution of monthly active rice area, which also reached its maximum in August.

Globally, Thailand had the highest estimated annual emission at 4.17 Mt CH_4, followed by India (4.08 Mt), China (3.35 Mt), and Indonesia (3.03 Mt). Thailand rice production was harvested from less than a quarter of the acreage of India and half that of China, meaning that emissions are driven by high emissions factors in Thailand. In a subsequent release, we will be implementing a new emissions factor estimation method to help interrogate whether the regional and seasonal emissions factors models overestimate emissions in Thailand and make updates as appropriate. Bangladesh contributed 1.83 Mt CH_4, while Vietnam contributed 1.55 Mt CH_4. Brazil, Tanzania, and Madagascar had lower annual emissions of approximately 0.529, 0.293, and 0.294 Mt CH_4, respectively.

The country-level annual totals reflect both the spatial extent and temporal characteristics of rice cultivation. Countries with multiple rice-growing seasons can maintain active rice areas and associated emissions over much of the year, whereas countries dominated by a single seasonal crop show more concentrated emission periods.

Overall, the 2025 results demonstrate that the framework provides both a global annual CH_4 inventory (23.97 Mt CH_4) and a temporally explicit representation of when those emissions occur. The monthly estimates retain the seasonal information derived from the rice cropping patterns rather than treating annual rice emissions as uniformly distributed throughout the year.

3.3 Validation

The rice extent, cropping-intensity and calendar maps underlying this inventory are validated in the peer-reviewed publications on which the method is based. Fatchurrachman et al. (2022) validated the 10 m Sentinel-1/-2 phenology-based mapping over Peninsular Malaysia against very-high-resolution and Google Street View reference points, reporting an overall accuracy of 95.95% and a kappa coefficient of 0.92, outperforming the NESEA-Rice10 product (90.27%, 0.80). Ginting et al. (2025) validated the LUCK-PALM cropping-intensity maps across nine Southeast

Asian countries at the pixel level (n = 58,885), reporting an overall accuracy of 98.3%, a kappa of 0.870, and an F1 score of 0.879 for rice classification, with per-country overall accuracy exceeding 96% (Malaysia 99.7%, Thailand 96%) and strong agreement with national harvested-area statistics. Ginting et al. (2026) extend this validation to cropping pattern, calendar and ecosystem type. Ginting et al. (2025) benchmarked their satellite-derived harvested area against FAOSTAT: the national total (42.99 Mha) agreed with FAOSTAT 2021 (45.09 Mha) to within 4.3% (R^2 = 0.92), and sub-national comparisons against provincial and district agricultural statistics yielded R^2 values of 0.85–0.97**3.**

For external context, the 2025 global total of 23.97 Mt CH_4 is broadly consistent with independent estimates of global rice-cultivation methane, which span roughly 25–30 Mt CH_4 per year across inventories and atmospheric studies. It sits below FAOSTAT's Tier-1 global total, consistent with the satellite model's more conservative detection of harvested area in fragmented and smallholder landscapes (Ginting et al., 2025). A full country-by-country comparison against FAOSTAT and EDGAR is planned for a future release.

4. Conclusion

This methodology produces a spatially explicit, monthly global inventory of methane emissions from rice cultivation spanning 2015–2025. The inventory covers 118 rice-producing countries, together representing approximately 98% of global rice production. For the 57 of these countries that are independently satellite-modeled — which account for roughly 97.6% of the inventory's total rice-cultivation methane — rice extent, cropping intensity, and seasonal calendars are resolved directly from 10 m Sentinel-1 and Sentinel-2 observations. Earlier years, and countries not resolved by satellite, are completed with a FAOSTAT harvested-area-indexed back-cast and, where no modeling exists, with the FAOSTAT rice-cultivation inventory, so that the dataset forms a consistent 2015–2025 series while keeping the three estimation methods explicit (Table S1). Relative to national inventories, the approach adds sub-national spatial detail, native monthly timing, and emission factors that reflect flooded, rainfed, and upland systems. Known limitations remain—the reliance on activity-based reconstruction for historical years, the sensitivity of a few countries (notably Thailand) to regional emission factors, and the conservative treatment of African rice under Tier-1 factors—and these are the focus of planned refinements, including an updated emission-factor model and extension of direct satellite modeling to additional countries and years.

Supplementary materials

Table S1 Data source for each modeled country and year. *10m* = Sentinel-1A/-2A/B modeling (independently modeled emissions). BC = a FAOSTAT harvested-area-indexed back-cast, anchored to each country's first 10 m-modeled year, used for the years preceding 10 m modeling. A country with "N/A" for a given year, or any country not shown, is reported directly from the FAOSTAT rice-cultivation inventory. The MODIS approach previously used for earlier years has been retired.

Country	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Bangladesh	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Bhutan	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*
Brazil	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Bulgaria	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Burkina Faso	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Burundi	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Cambodia	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Cameroon	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
China	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Ecuador	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*
Egypt	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*
Ethiopia	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*

Country	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
France	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Gambia	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Greece	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Guinea	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Guinea-Bissau	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Guyana	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*
India	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Indonesia	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Iran (Islamic Republic of)	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Italy	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Japan	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Kenya	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Korea (the Democratic People's Republic of)	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Korea (the Republic of)	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Lao People's Democratic Republic (the)	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Madagascar	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*

Country	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Malaysia	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Mali	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Mauritania	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Mexico	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*
Morocco	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Myanmar	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Nepal	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Niger	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Nigeria	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Pakistan	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Peru	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*
Philippines (the)	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Portugal	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Russia	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*
Rwanda	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Senegal	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*

Country	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Sierra Leone	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
Spain	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Sri Lanka	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Suriname	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*
Taiwan (Province of China)	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Thailand	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Timor-Leste	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*
Turkey	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*
Uganda	BC	BC	BC	BC	BC	BC	BC	BC	BC	BC	*10m*
United Republic of Tanzania	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
United States of America (the)	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*
Uruguay	BC	BC	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*
Viet Nam	BC	BC	BC	BC	BC	BC	*10m*	*10m*	*10m*	*10m*	*10m*

Table S2 Seasonally integrated methane (CH4) emission factors (EFs) in various conditions and locations of the world that were used in this study. Mean emission factors and standard deviation (SD) are provided.

Country	ISO3 country	Mean CH_4 Emission (kg CH_4/ha/season)	SD CH_4 Emission (kg CH_4/ha/season)	References
Bangladesh	BGD	168.2	80.4	(Islam *et al.*, 2020)
Bhutan	BTN	81	42.5	India EF
Brazil	BRA	430.1	149.6	(Camargo *et al.*, 2018; Zschornack *et al.*, 2018)
Bulgaria	BGR	292	116	Italy EF
Burkina Faso	BFA	183.6	51.04	Africa EF (Mboyerwa, 2022)
Burundi	BDI	183.6	51.04	Africa EF (Mboyerwa, 2022)
Cambodia	KHM	145.3	31	(Vibol and Towprayoon, 2010)
Cameroon	CMR	183.6	51.04	Africa EF (Mboyerwa, 2022)
Ecuador	ECU	430.1	149.6	Brazil EF
Egypt	EGY	183.6	51.04	(Mboyerwa, 2022)
Ethiopia	ETH	183.6	51.04	(Mboyerwa, 2022)
France	FRA	292	116	Italy EF
Gambia	GMB	183.6	51.04	Africa EF (Mboyerwa, 2022)
Greece	GRC	292	116	Italy EF
Guinea	GIN	183.6	51.04	Africa EF (Mboyerwa, 2022)
Guinea-Bissau	GNB	183.6	51.04	Africa EF (Mboyerwa, 2022)

Country	ISO3 country	Mean CH_4 Emission (kg CH_4/ha/season)	SD CH_4 Emission (kg CH_4/ha/season)	References
Guyana	GUY	430.1	149.6	Brazil EF
Hungary	HUN	292	116	Italy EF
India	IND	81	42.5	(Bhatia *et al.*, 2005; Kritee *et al.*, 2018; Oo *et al.*, 2018)
Indonesia	IDN	339.8	102.1	(Setyanto *et al.*, 2018)
Iran (Islamic Republic of)	IRN	81	42.5	India EF
Italy	ITA	292	116	(Lagomarsino *et al.*, 2016; Mazza *et al.*, 2016; Meijide *et al.*, 2017)
Japan	JPN	469.8	302.4	(Camargo *et al.*, 2018; Toma *et al.*, 2019)
Kenya	KEN	183.6	51.04	Africa EF (Mboyerwa, 2022)
Korea (the Democratic People's Republic of)	PRK	349.4	93	Korea (the Republic of) EF
Korea (the Republic of)	KOR	349.4	93	(Gutierrez, Kim and Kim, 2013; Lim *et al.*, 2021)
Lao People's Democratic Republic (the)	LAO	78.3	31.6	(Maneepitak *et al.*, 2019)
Madagascar	MDG	183.6	51.04	Africa EF (Mboyerwa, 2022)
Malawi	MWI	183.6	51.04	Africa EF (Mboyerwa, 2022)
Malaysia	MYS	178.3	118.5	(Fazli and Man, 2014)
Mali	MLI	183.6	51.04	Africa EF (Mboyerwa, 2022)

Country	ISO3 country	Mean CH_4 Emission (kg CH_4/ha/season)	SD CH_4 Emission (kg CH_4/ha/season)	References
Mauritania	MRT	183.6	51.04	Africa EF (Mboyerwa, 2022)
Mexico	MEX	202	121.9	USA EF
Morocco	MAR	183.6	51.04	Africa EF (Mboyerwa, 2022)
Myanmar	MMR	30.1	12.5	(Win *et al.*, 2020)
Nepal	NPL	81	42.5	India EF
Niger	NER	183.6	51.04	Africa EF (Mboyerwa, 2022)
Nigeria	NGA	183.6	51.04	Africa EF (Mboyerwa, 2022)
North Macedonia	MKD	292	116	Italy EF
Pakistan	PAK	81	42.5	India EF
Peru	PER	430.1	149.6	Brazil EF
Philippines (the)	PHL	258	192.7	(Alberto *et al.*, 2014; Sander, Samson and Buresh, 2014; Sibayan *et al.*, 2018)
Portugal	PRT	405.7	202.9	Spain EF
Russia	RUS	81	42.5	India EF
Rwanda	RWA	183.6	51.04	Africa EF (Mboyerwa, 2022)
Senegal	SEN	183.6	51.04	Africa EF (Mboyerwa, 2022)
Sierra Leone	SLE	183.6	51.04	Africa EF (Mboyerwa, 2022)
Spain	ESP	405.7	202.9	(Moreno-García, Guillén and Quílez, 2020;

Country	ISO3 country	Mean CH_4 Emission (kg CH_4/ha/season)	SD CH_4 Emission (kg CH_4/ha/season)	References
				Martínez-Eixarch *et al.*, 2021)
Sri Lanka	LKA	81	42.5	India EF
Suriname	SUR	430.1	149.6	Brazil EF
Taiwan (Province of China)	TWN	112	91.4	(Chang, 2001)
Timor-Leste	TLS	339.8	102.1	(Setyanto *et al.*, 2018)
Turkey	TUR	81	42.5	India EF
Uganda	UGA	183.6	51.04	Africa EF (Mboyerwa, 2022)
United Republic of Tanzania	TZA	183.6	51.04	Africa EF (Mboyerwa, 2022)
United States of America (the)	USA	202	121.9	(Hatala *et al.*, 2012; Humphreys *et al.*, 2019; Della Lunga *et al.*, 2021; Karki *et al.*, 2021)
Uruguay	URY	430.1	149.6	Brazil EF

Table S3 summarizes emissions factors and their standard deviation for five regions in China (Sun 2020). For regions where it is common to have multiple rice harvests, unique emissions factors were provided to help illustrate seasonal variation. These emissions factors were applied to modeled harvested area estimates to characterize annual methane emissions.

Table S3 China subnational EFs reported in Sun (2020)

Region	Season	Mean (kg CH4/ha)	Standard Deviation
South China	Early Season	50.5	83.41
	Late-rice	182.3	156.65
	All Rice	116.4	146.14
Southwest China	Single Rice	244	220.36
	All Rice	244	220.36
Yangtze River	Early Season	99.2	140.68
	Late-rice	224.8	224.03
	Single Rice	188.5	173.32
	All Rice	174	188.75
Northeast	Single Rice	74.4	133.62
Huang-Huai-Hai	Single Rice	43.2	15.41

Table S4 Thailand subnational estimated seasonal rice field methane rates. Major and second refers to “wet season rice cropping” and “dry season rice cropping”, respectively (Katoh, 1999). Table modified from Katoh (1999). Blank cells indicate no value given. Asterisk with numbers refer to citations- *1 = Yagi et al. (1994), *2 = Katoh et al. (1999a), and *3 = Katoh et al. (1999b). A season with a “N/A” indicates no value provided.

Site	Year	Rice cultivation	Flooding period (day)	CH4 flux (mg m-2 hr-1)	Estimated seasonal emission (g m-2 season-1)	
					Second	Major

Site	Year	Rice cultivation	Flooding period (day)	CH4 flux (mg m-2 hr-1)	Estimated seasonal emission (g m-2 season-1)	
					Second	Major
Khon kaen	1991	Major *1	97	16.4	N/A	50.8
		Second *1	109	19.4	38.2	N/A
Khlong Lugang	1991	Second *1	83	3.1	6.1	N/A
Chai Net	1991	Major *1	94	1.1	N/A	2.5
Bang Khen	1992	Major *2	106	21.8	N/A	55.5
		Second *2	120	4.3	12.4	N/A
	1994	Second	118	6.7	19	N/A
Phitsanulok	1992	Major *3	98	7.4	N/A	17.4
	1993	Second *3	113	6.6	17.9	N/A
San Pa Thong	1993	Major *3	103	16.1	N/A	39.8
	1994	Second *3	101	8.8	21.3	N/A
Phtae	1993	Major *3	128	22.2	N/A	68.2
	1994	Second *3	127	15.9	48.5	N/A
Khon Kaen	1994	Major *3	129	19.8	N/A	61.3
	1995	Second *3	96	15.1	34.8	N/A
Surin	1994	Major *3	123	13.3	N/A	39.3
	1995	Second*3	120	15.4	44.4	N/A
Mean					***26.9***	***41.8***

Table S5 Vietnam subnational emission factors reported in Vo et al. (2020). In each of these regions, rice production involved multiple harvests. Unique emissions factors were provided to help illustrate seasonal variation in emissions across successive harvests. These emissions factors were applied to modeled harvested area estimates to characterize annual methane emissions.

Region of Vietnam	Season	Average emissions (kg CH4 ha^{-1} $season^{-1}$)	Standard Deviation
North	Early	271	150
	Late	404	173
Central	Early	321	237
	Middle	321	237
South	Early	174	82
	Middle	277	116
	Late	356	481

Table S6 Monthly activity rice area (ha) for 2025.

No	ISO3	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	IND	6,191,981	5,727,079	5,497,200	5,208,417	-	3,133,684	41,161,348	44,149,033	44,149,033	44,613,935	24,205,994	1,468,474
2	CHN	0	0	2,449,905	2,449,905	13,461,475	19,378,554	19,412,393	19,412,393	17,010,840	6,877,351	0	0
3	THA	1,935,179	1,775,340	1,607,189	2,518,215	7,533,351	9,341,452	9,821,523	9,310,758	8,776,034	9,033,028	4,573,695	1,313,252
4	BGD	4,292,375	4,557,626	4,557,626	4,532,882	1,010,698	852,659	2,613,780	5,676,942	5,676,942	5,676,942	5,650,873	2,278,356
5	MMR	2,815,085	2,815,085	2,398,097	555,291	1,304,986	1,016,434	4,455,071	5,718,855	5,513,136	5,190,841	6,580,515	3,502,909
6	IDN	3,350,940	3,811,862	3,530,552	3,792,086	3,450,072	3,988,961	3,890,971	3,694,463	2,336,844	1,382,711	1,755,588	2,561,625
7	KHM	1,013,704	1,013,704	460,261	332,496	1,846,919	1,941,132	2,972,010	2,972,010	2,814,061	2,886,203	1,954,516	1,195,688
8	VNM	1,536,001	2,076,799	2,060,603	1,928,650	2,134,926	2,095,697	2,133,889	1,815,105	1,495,946	1,146,680	942,000	1,468,997
9	PAK	0	0	0	0	0	1,953,998	3,379,400	3,379,400	3,379,400	3,379,400	522,665	0
10	PHL	1,163,181	1,011,034	793,863	579,606	415,958	1,130,209	2,145,911	2,054,159	1,970,471	1,597,422	573,204	869,586
11	TZA	1,482,363	1,596,367	1,596,367	1,596,367	705,261	66,234	0	0	0	0	0	1,154,852
12	MDG	1,294,762	1,419,935	1,360,232	1,360,232	595,026	47,773	47,773	51,062	147,124	147,124	232,103	1,203,293
13	JPN	0	0	0	0	916,447	1,814,609	1,814,609	1,814,609	1,359,970	0	0	0
14	BRA	1,389,334	1,389,334	380,803	0	0	0	0	0	0	209,859	1,389,334	1,389,334
15	NPL	0	0	0	0	0	606,855	1,379,191	1,379,191	1,379,191	1,379,191	0	0
16	LAO	50,404	50,404	50,404	50,404	309,061	718,041	1,072,514	1,072,514	1,072,514	1,072,514	407,040	50,404
17	USA	0	0	0	228,592	1,097,120	1,097,120	1,097,120	915,581	868,528	0	0	0
18	LKA	595,521	379,426	201,901	581,958	595,521	595,521	535,710	13,563	0	217,605	521,008	595,521
19	NGA	78,615	162,696	204,123	282,738	282,738	376,758	708,767	662,911	662,911	662,911	484,810	78,615
20	KOR	0	0	0	0	548,146	673,913	673,913	673,913	673,913	0	0	0
21	IRN	0	0	0	474,863	325,038	474,863	474,863	474,863	149,825	149,825	0	0
22	PRK	0	0	0	0	322,807	535,181	535,181	535,181	535,181	0	0	0
23	MYS	197,700	167,162	102,263	106,184	168,686	222,718	229,039	212,491	215,660	122,307	222,641	209,115
24	MLI	34,149	34,149	34,149	34,149	0	64,284	201,132	333,931	471,852	471,852	270,720	137,921
25	EGY	0	0	0	41,594	313,612	313,612	313,612	313,612	313,612	0	0	0
26	GIN	0	0	0	0	0	0	43,900	221,003	232,806	232,806	220,917	36,913
27	PER	90,409	90,409	127,323	139,387	152,617	87,240	84,073	72,009	58,779	33,747	33,747	0
28	TWN	0	0	137,646	146,922	145,968	99,305	37,394	98,807	104,275	104,275	76,157	5,468

No	ISO3	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
29	ECU	0	82,174	82,174	95,968	118,437	91,252	139,299	125,505	103,036	90,227	0	0
30	ITA	0	0	0	8,718	179,504	179,650	179,650	179,650	170,932	146	0	0
31	TUR	0	0	0	7,445	117,443	117,667	117,667	117,667	110,222	224	0	0
32	GUY	73,519	73,519	37,422	0	36,097	73,519	73,519	73,519	37,422	0	36,097	73,519
33	SEN	3,937	38,457	38,457	34,520	34,520	38,457	13,071	79,602	79,602	79,602	70,468	70,468
34	MRT	3,412	42,543	42,543	39,130	39,130	42,543	8,439	67,816	67,816	67,816	62,789	62,789
35	RUS	0	0	0	0	85,626	85,626	85,626	85,626	85,626	0	0	0
36	ESP	0	0	0	358	67,625	67,743	67,743	67,743	67,385	118	0	0
37	URY	63,944	63,944	55,125	0	0	0	0	0	0	8,819	63,944	63,944
38	NER	8,551	11,595	11,595	11,595	5,128	0	11,398	43,530	56,378	56,378	54,231	23,157
39	SLE	0	0	0	0	0	0	45,288	58,183	58,183	58,183	58,183	12,895
40	UGA	12,006	12,006	12,006	30,402	30,402	21,850	30,402	30,402	30,402	26,949	23,801	18,396
41	GNB	0	0	0	0	0	0	12,086	46,563	46,563	46,563	46,563	34,477
42	GRC	0	0	0	15,025	38,940	38,951	38,951	38,951	23,926	12	0	0
43	SUR	20,934	20,934	10,043	0	10,890	20,934	20,934	20,934	10,043	0	10,890	20,934
44	ETH	0	0	0	0	0	0	10,451	28,976	28,976	28,976	28,976	18,525
45	BFA	0	3,083	3,083	3,083	3,083	6,373	12,769	23,906	23,906	23,906	15,866	0
46	PRT	0	0	0	241	23,210	23,263	23,263	23,263	23,023	53	0	0
47	CMR	0	1,996	11,091	11,091	11,091	9,096	10,191	14,001	14,001	14,001	6,711	0
48	KEN	799	5,616	5,616	5,616	5,616	5,616	10,357	11,396	11,396	12,195	12,195	1,838
49	TLS	0	649	2,031	15,051	15,051	15,051	14,402	13,020	0	0	0	0
50	BTN	0	0	0	14	0	4	14,258	14,506	14,120	14,121	14,117	4
51	RWA	198	6,395	6,395	6,395	6,395	6,395	0	5,853	6,051	6,051	6,051	6,051
52	BGR	0	0	0	3,992	12,263	12,264	12,264	12,264	8,272	0	0	0
53	BDI	3,552	6,367	4,959	4,959	4,959	3,936	1,024	3,919	5,040	6,448	6,448	5,425
54	MEX	4,493	7,293	7,293	7,293	5,967	3,328	4,097	4,097	4,097	4,097	3,569	1,326
55	MKD	0	0	0	2,051	4,982	4,982	4,982	4,982	2,931	0	0	0
56	GMB	0	0	0	0	0	0	740	3,988	3,988	3,988	3,988	

No	ISO3	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
													0
57	FRA	0	0	0	52	2,351	2,351	2,351	2,351	2,300	0	0	0
58	MAR	0	0	0	0	935	935	935	935	35	0	0	0

Table S7 Monthly Emission CH_4 (tonnes) from rice cultivation for 2025.

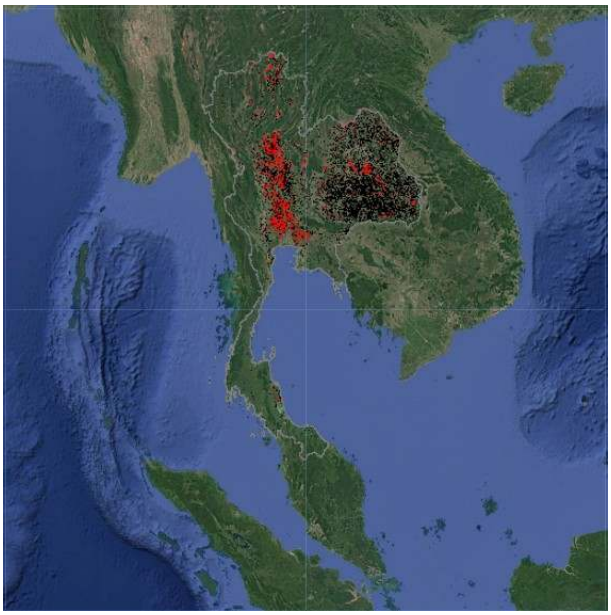
No	ISO3	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
1	THA	107,547	103,693	102,158	136,149	476,515	588,264	615,988	589,374	565,624	572,584	260,046	48,641	4,166,582
2	IND	123,726	114,312	109,657	104,979	0	50,764	738,854	797,878	797,878	807,292	405,565	26,602	4,077,508
3	CHN	0	0	93,441	93,441	392,167	620,162	659,647	659,647	546,953	282,779	0	0	3,348,237
4	IDN	275,506	310,501	285,814	297,700	269,964	315,110	310,945	295,043	195,429	116,828	143,752	208,455	3,025,047
5	BGD	173,209	184,362	184,362	183,322	41,911	33,519	92,759	214,746	214,746	214,746	213,869	82,530	1,834,082
6	VNM	131,526	170,101	161,185	139,602	150,615	155,428	160,350	130,089	100,360	68,671	60,100	117,357	1,545,386
7	PHL	72,780	62,967	48,960	35,140	25,409	69,038	133,602	127,684	122,286	98,225	36,024	55,264	887,379
8	JPN	0	0	0	0	96,785	202,270	202,270	202,270	148,875	0	0	0	852,470
9	KHM	28,624	28,624	12,541	9,662	49,049	51,331	80,762	80,762	76,172	78,268	55,646	34,335	585,775
10	BRA	119,504	119,504	32,755	0	0	0	0	0	0	18,051	119,504	119,504	528,823
11	MDG	47,164	53,791	50,716	50,716	22,619	1,754	1,754	1,875	5,402	5,402	9,406	43,806	294,406
12	TZA	52,816	57,440	57,440	57,440	24,721	2,432	0	0	0	0	0	40,790	293,080
13	PAK	0	0	0	0	0	31,653	58,398	58,398	58,398	58,398	8,466	0	273,711
14	MMR	17,422	17,422	14,912	3,343	8,166	6,428	27,034	35,171	33,623	31,683	40,048	21,522	256,773
15	KOR	0	0	0	0	38,302	49,287	49,287	49,287	49,287	0	0	0	235,451
16	USA	0	0	0	11,068	46,155	46,155	46,155	36,987	35,087	0	0	0	221,607
17	PRK	0	0	0	0	22,557	41,107	41,107	41,107	41,107	0	0	0	186,984
18	NGA	2,405	6,263	8,946	11,832	11,832	14,511	26,019	24,336	23,855	23,855	17,317	2,405	173,575
19	NPL	0	0	0	0	0	9,830	25,469	25,469	25,469	25,469	0	0	111,707
20	LKA	12,053	7,677	3,815	11,510	11,785	11,785	10,574	275	0	4,400	10,544	12,053	96,469
21	ECU	0	7,928	7,928	9,411	11,827	8,904	14,978	13,494	11,079	9,701	0	0	95,252
22	MLI	1,567	1,567	1,567	1,567	0	2,360	8,641	14,735	21,065	21,065	12,424	6,330	92,889
23	PER	7,777	7,777	11,746	13,043	14,466	9,380	8,314	7,017	5,594	2,903	2,903	0	90,919

No	ISO3	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
24	MYS	8,160	6,836	3,943	4,231	6,552	9,081	9,242	8,504	8,869	5,294	9,393	8,911	89,016
25	LAO	789	789	789	789	4,033	10,207	15,898	15,898	15,898	15,898	6,144	789	87,923
26	GUY	7,905	7,905	4,024	0	3,881	7,905	7,905	7,905	4,024	0	3,881	7,905	63,238
27	EGY	0	0	0	1,527	11,515	11,515	11,515	11,515	11,515	0	0	0	59,100
28	ITA	0	0	0	509	10,482	10,490	10,490	10,490	9,981	8	0	0	52,451
29	IRN	0	0	0	9,009	6,582	9,009	9,009	9,009	2,427	2,427	0	0	47,471
30	GIN	0	0	0	0	0	0	1,721	9,617	10,159	10,159	9,613	1,463	42,732
31	URY	5,500	5,500	4,742	0	0	0	0	0	0	759	5,500	5,500	27,501
32	ESP	0	0	0	29	5,487	5,496	5,496	5,496	5,467	10	0	0	27,482
33	TWN	0	0	3,807	4,067	4,040	2,734	1,047	2,766	2,920	2,920	2,132	153	26,586
34	SEN	145	1,412	1,412	1,267	1,267	1,412	564	3,006	3,006	3,006	2,587	2,587	21,672
35	MRT	125	1,562	1,562	1,437	1,437	1,562	356	2,536	2,536	2,536	2,305	2,305	20,260
36	SUR	2,251	2,251	1,080	0	1,171	2,251	2,251	2,251	1,080	0	1,171	2,251	18,007
37	NER	373	532	532	532	235	0	438	1,818	2,389	2,389	2,290	949	12,479
38	GRC	0	0	0	877	2,274	2,275	2,275	2,275	1,397	1	0	0	11,373
39	UGA	472	472	472	1,148	1,116	802	1,145	1,145	1,314	1,187	1,043	844	11,162
40	TUR	0	0	0	121	1,902	1,906	1,906	1,906	1,785	4	0	0	9,530
41	PRT	0	0	0	20	1,883	1,887	1,887	1,887	1,868	4	0	0	9,437
42	GNB	0	0	0	0	0	0	444	1,709	1,709	1,709	1,709	1,266	8,547
43	RUS	0	0	0	0	1,367	1,367	1,367	1,367	1,367	0	0	0	6,835
44	SLE	0	0	0	0	0	0	592	1,065	1,065	1,065	1,065	473	5,327
45	ETH	0	0	0	0	0	0	384	1,064	1,064	1,064	1,064	680	5,319
46	TLS	0	44	138	1,023	1,023	1,023	979	885	0	0	0	0	5,114
47	BFA	0	142	142	142	142	234	484	995	995	995	685	0	4,955
48	CMR	0	92	509	509	509	417	441	616	616	616	281	0	4,606
49	BGR	0	0	0	233	716	716	716	716	483	0	0	0	3,581
50	KEN	37	206	206	206	206	206	380	418	418	455	455	75	3,270
51	MEX	182	295	295	295	241	134	166	166	166	166	144	54	2,301
52	RWA	7	235	235	235	235	235	0	215	222	222	222	222	2,285

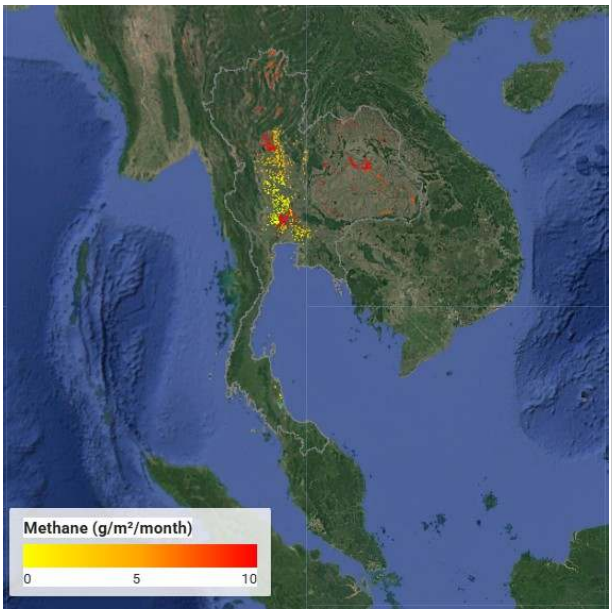
No	ISO3	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
53	BDI	130	234	182	182	182	145	38	144	185	237	237	199	2,094
54	MKD	0	0	0	120	291	291	291	291	171	0	0	0	1,455
55	BTN	0	0	0	0	0	0	232	237	229	229	229	0	1,157
56	GMB	0	0	0	0	0	0	27	176	176	176	176	0	732
57	FRA	0	0	0	3	137	137	137	137	134	0	0	0	687
58	MAR	0	0	0	0	34	34	34	34	34	0	0	0	172
Monthly totals =		**1,199,703**	**1,282,438**	**1,212,014**	**1,198,437**	**1,783,785**	**2,404,945**	**3,402,764**	**3,513,906**	**3,173,959**	**2,493,854**	**1,447,942**	**856,220**	**23,969,967**

Figure S1 illustrates spatial distribution of monthly active rice areas and CH_4 emission estimates in Thailand for 2025. In the active rice maps, red indicates active rice areas, whereas black indicates non-active rice areas.

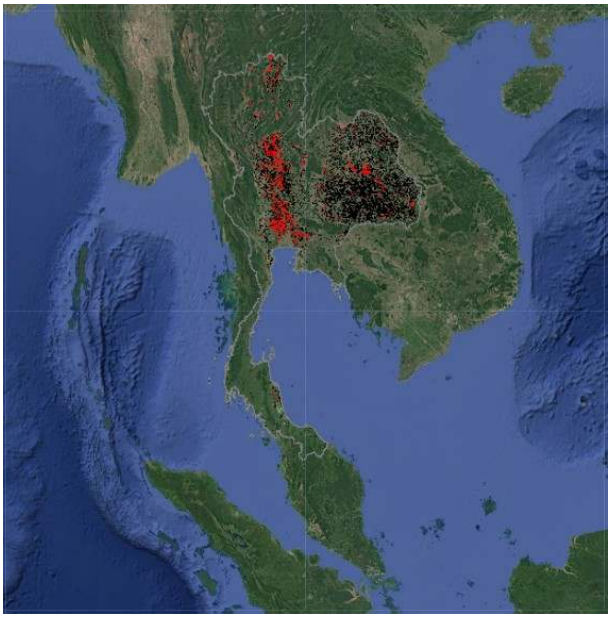
(a) Active rice area in Jan 2025



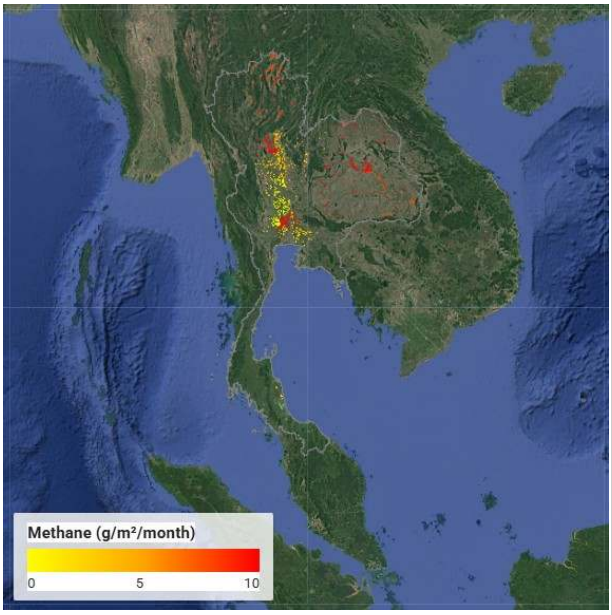
(b) CH4 emission in Jan 2025



(c) Active rice area in Feb 2025

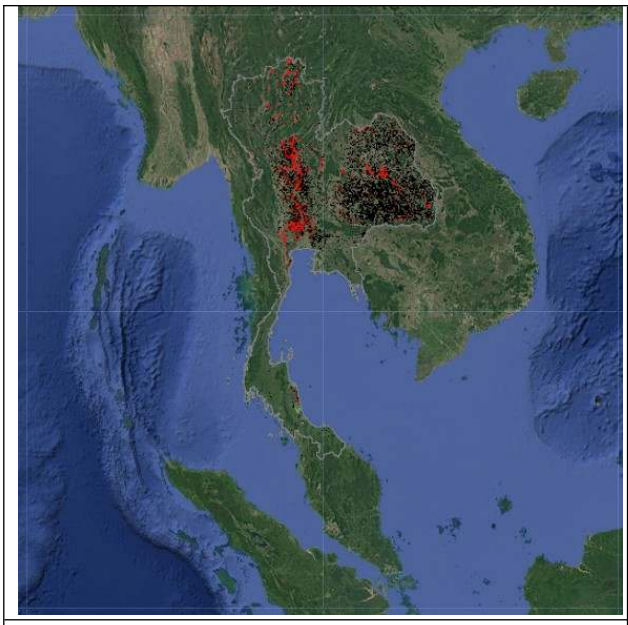


(d) CH4 emission in Feb 2025

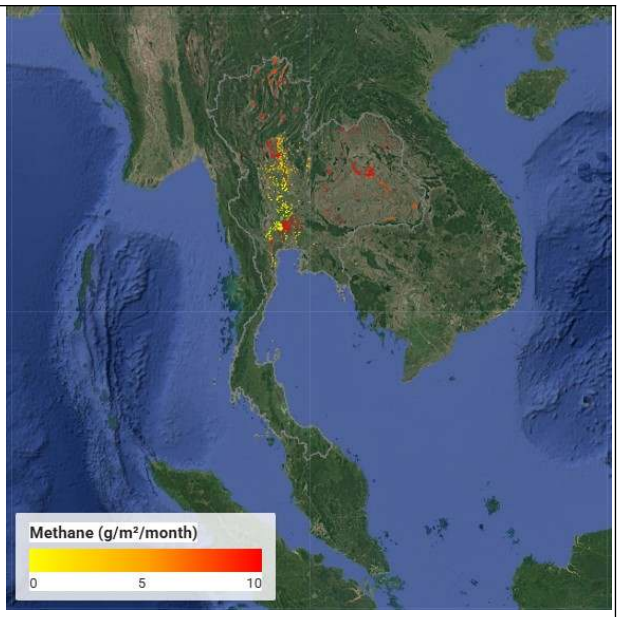


(e) Active rice area in Mar 2025

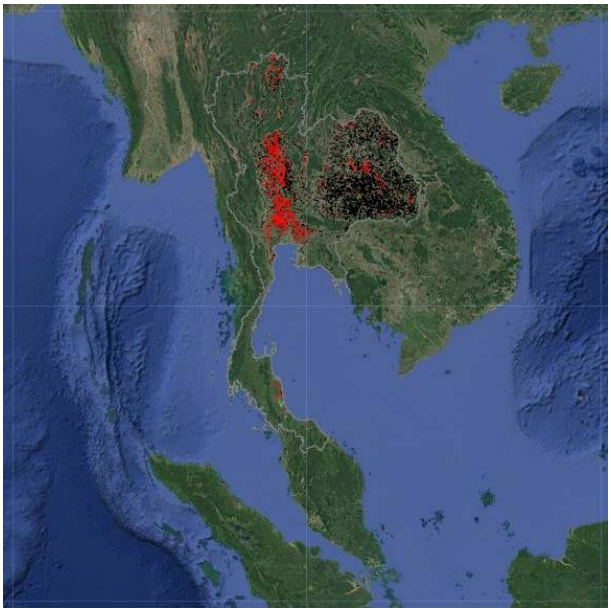
(f) CH4 emission in Mar 2025



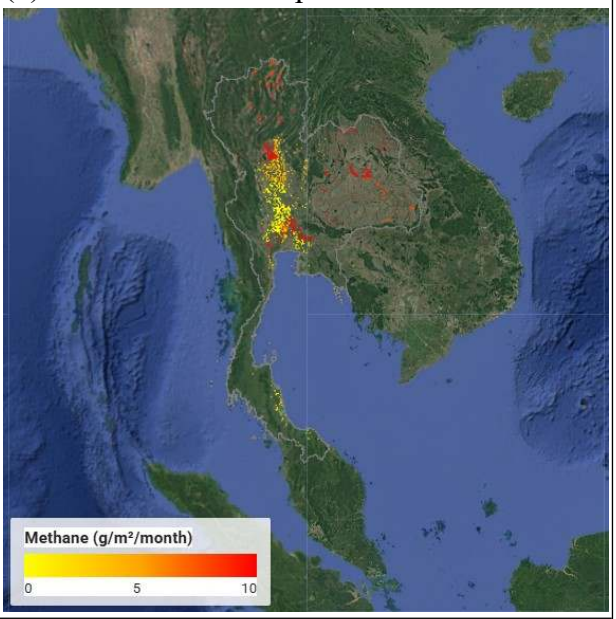
(g) Active rice area in Apr 2025



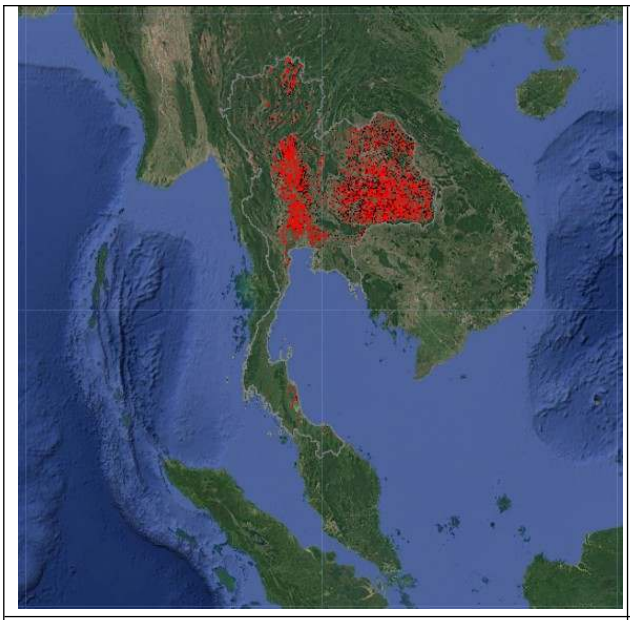
(h) CH4 emission in Apr 2025



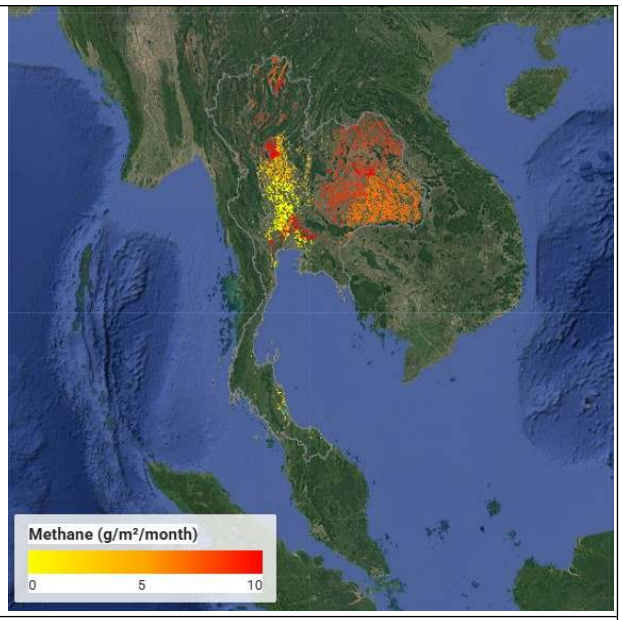
(i) Active rice area in May 2025



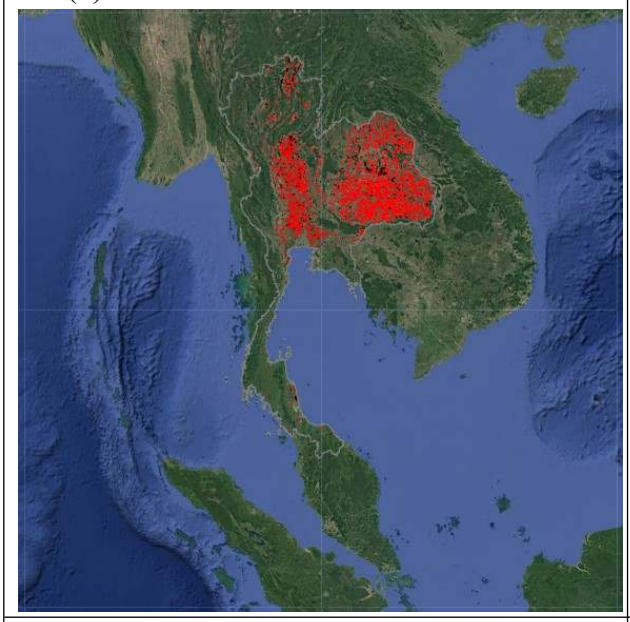
(j) CH4 emission in May 2025



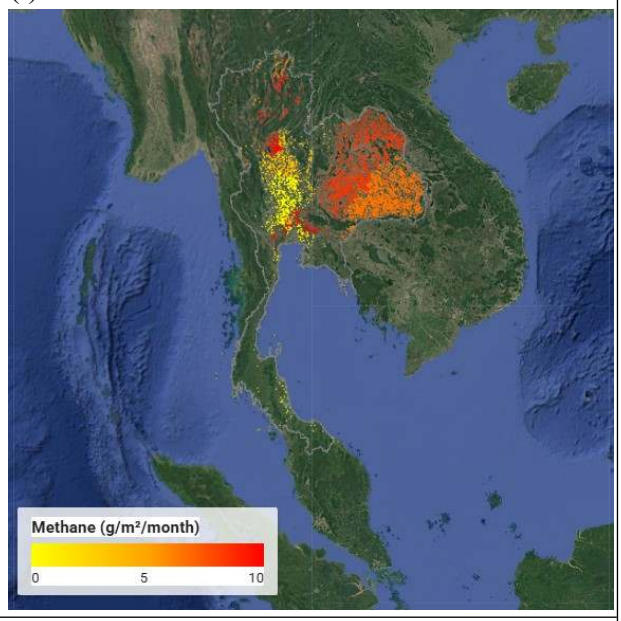
(k) Active rice area in Jun 2025



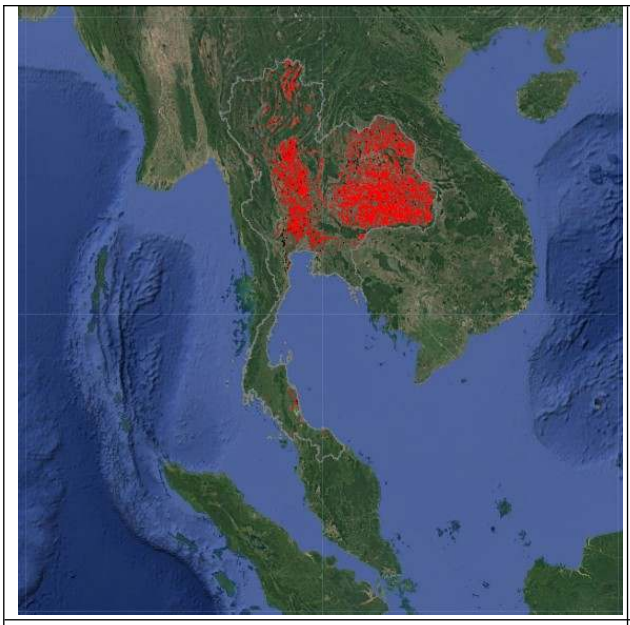
(l) CH4 emission in Jun 2025



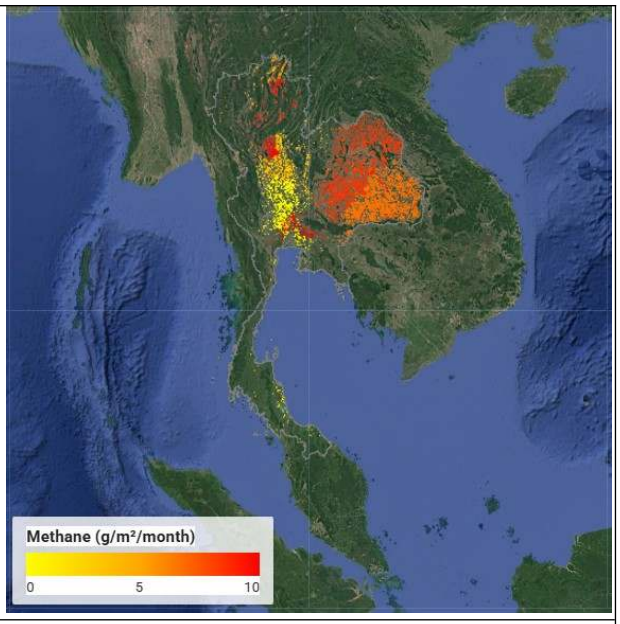
(m) Active rice area in Jul 2025



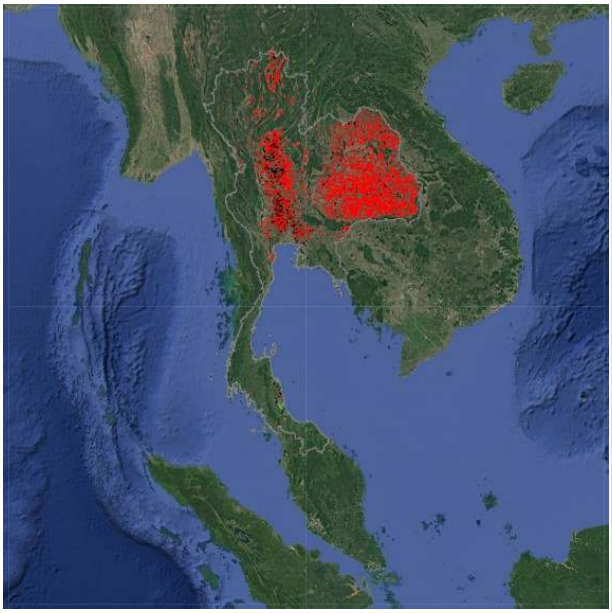
(n) CH4 emission in Jul 2025



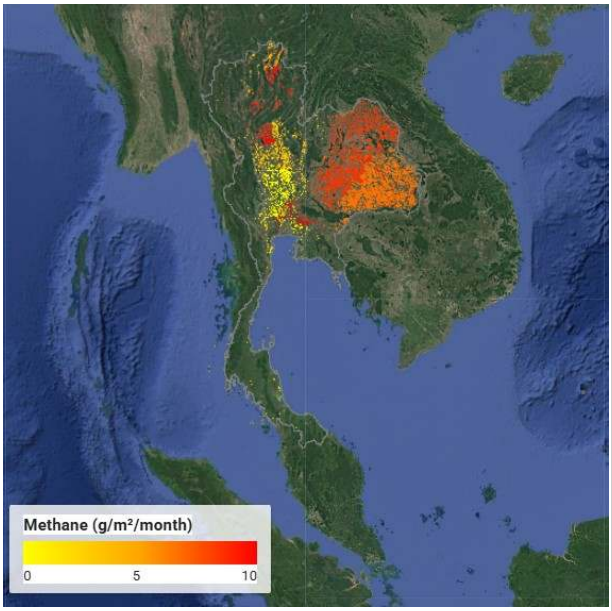
(o) Active rice area in Aug 2025



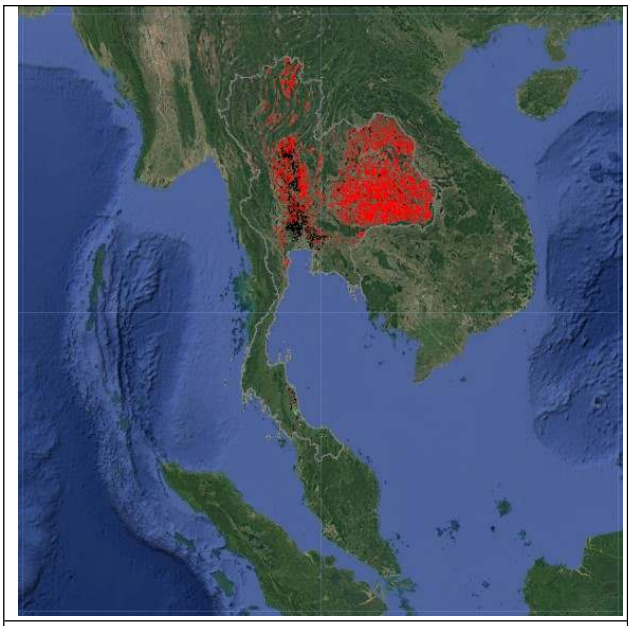
(p) CH4 emission in Aug 2025



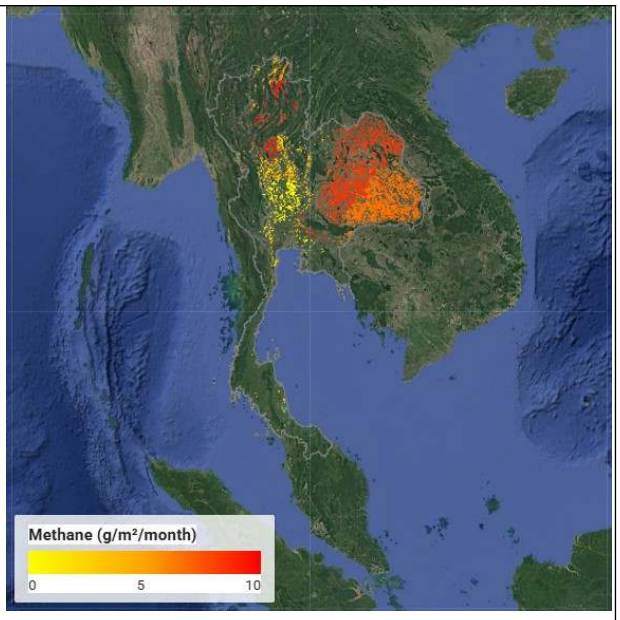
(q) Active rice area in Sep 2025



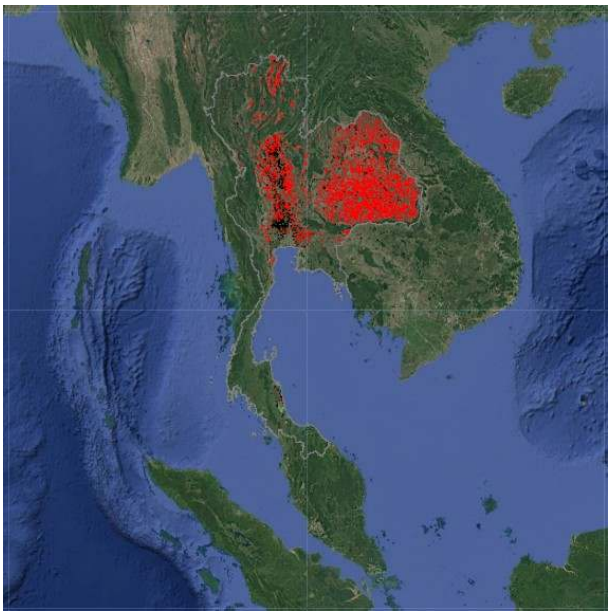
(r) CH4 emission in Sep 2025



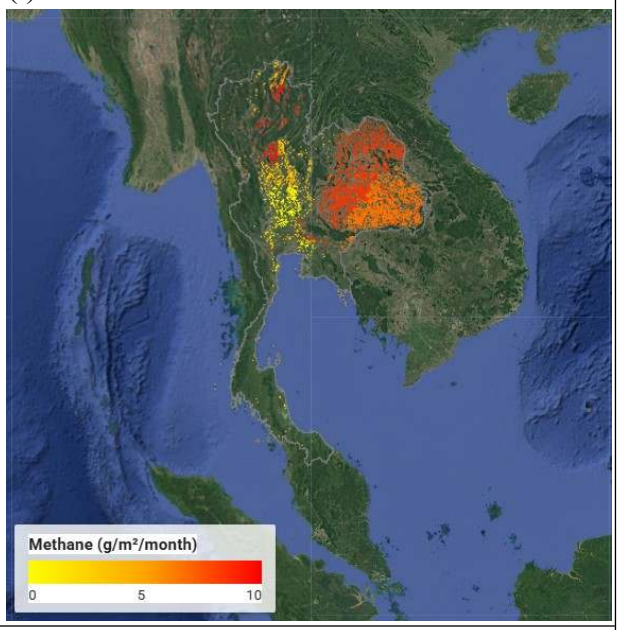
(s) Active rice area in Oct 2025



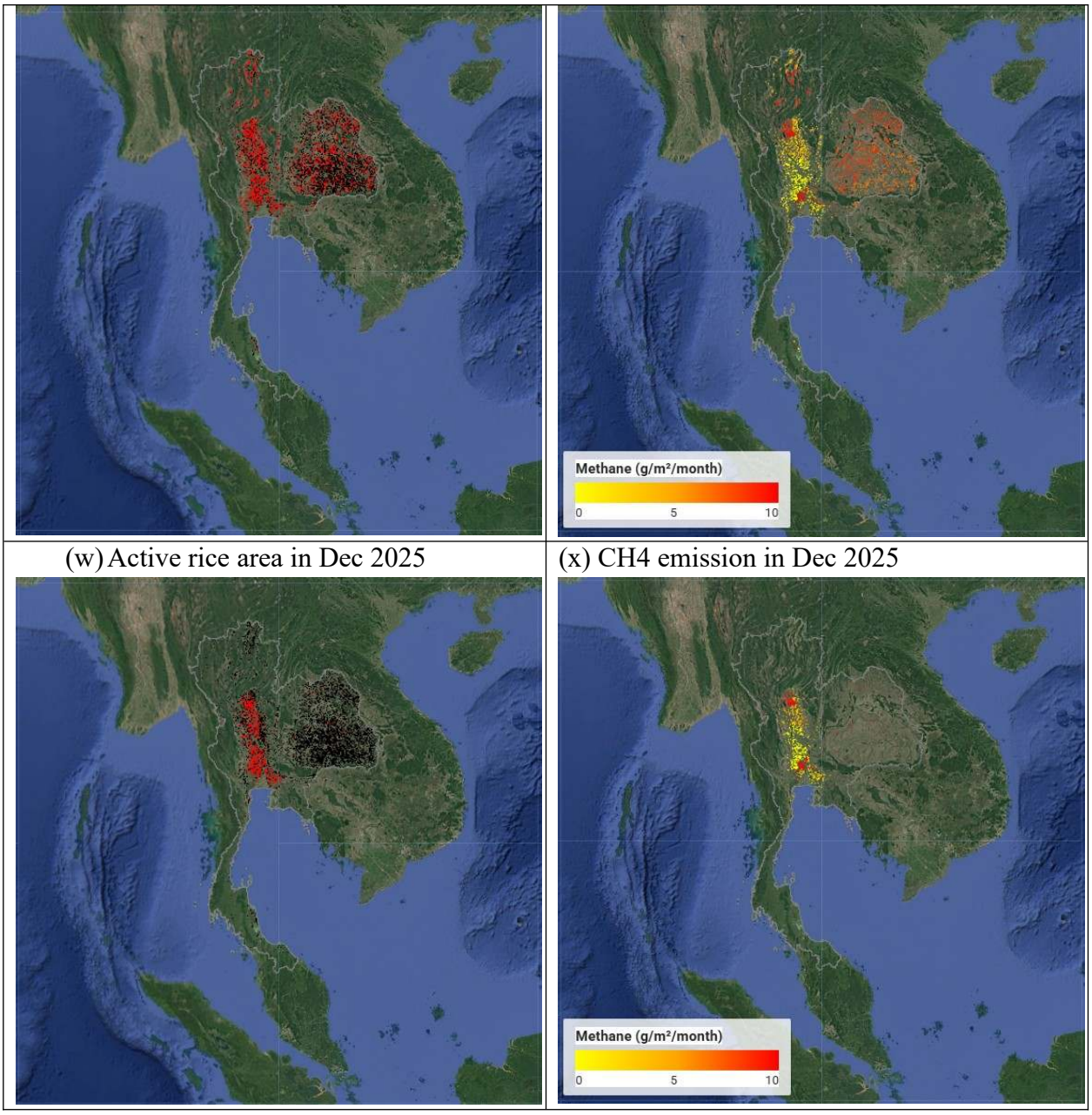
(t) CH4 emission in Oct 2025



(u) Active rice area in Nov 2025



(v) CH4 emission in Nov 2025



(w) Active rice area in Dec 2025 (x) CH4 emission in Dec 2025

Metadata

The Agriculture sector: "Rice Cultivation Methane Emissions Estimated from Satellite, Statistical, and Inventory-Based Methods" reports the following data on the Climate TRACE website:

- Country-level CH_4, and 20 and 100 year GWPs emissions from rice cultivation.

Emissions estimates were reported for years 2015 to 2025, with earlier years provided by a FAOSTAT harvested-area-indexed back-cast for modeled countries, or directly from the FAOSTAT rice-cultivation inventory for countries not modeled (see Table S1). Other approaches to estimate rice emissions are in the Climate TRACE GitHub methodology repository. The data

generated here has been combined with the other approaches to estimate rice cultivation emissions globally. All data is freely available on the Climate TRACE website (https://climatetrace.org/). A detailed description of what is available is described in Table S8. Note, uncertainty definitions are defined in Tables S2, S3, S4, and S5, the standard deviation associated with each country. Confidence definitions are in the tables below.

Table S8 Metadata for Rice Cultivation Emissions Estimates.

General Description	Definition
Sector definition	*Country-level rice cultivation emissions*
UNFCCC sector equivalent	*3.C.7 Rice Cultivation*
Temporal Coverage	*2015 – 2025 (see Table S1)*
Temporal Resolution	*Monthly (on website, see Temporal Disaggregation of Emissions Data for the Climate TRACE Inventory)*
Data format	*CSV*
Coordinate Reference System	*None. ISO3 country code provided*
Number of assets/countries available for download	*60 countries (2025; see Table S1), of which 57 are modeled from satellite observation. The remaining countries use FAOSTAT data.*
Ownership	*Country*
What emission factors were used?	*Country and subnational literature values (Tables S2–S5)*
What is the difference between a "0" versus "NULL/none/nan" data field?	*"0" values are for true non-existent emissions. If we know that the sector has emissions for that specific gas, but the gas was not modeled, this is represented by "NULL/none/nan"*
total_CO2e_100yrGWP and total_CO2e_20yrGWP conversions	*Climate TRACE uses IPCC AR6 CO_2e GWPs. CO_2e conversion guidelines are here: https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_FullReport_small.pdf*

Confidence values are based on the level of information detail used to estimate emissions, which is defined as the following:

- Subgroup 1: 10m resolution, regional EF
- Subgroup 2: 10m resolution, country level EF
- Subgroup 3: 10m resolution, Tier 1 EF or EF from neighboring country
- Subgroup 4: FAOSTAT-indexed back-cast, country level EF
- Subgroup 5: FAOSTAT-indexed back-cast, Tier 1 EF or EF from neighboring country
- Subgroup 6: backfilled by Climate TRACE

Low to very high confidence values for each group are shown in Table S9 and country specific confidences are shown in Table S10.

Table S9 Confidence definitions for rice cultivation emissions by subgroup number.

Data attributes	Definitions	Subgroup 1	Subgroup 2	Subgroup 3	Subgroup 4	Subgroup 5	Subgroup 6
type	Not used; NA						
capacity_description	Not used; NA						
capacity_factor_description	Not used; NA						
activity_description	Rice field that cultivated by rice	High	High	High	Medium	Medium	Low
CO2_emissions_factor	Not used; NA						
CH4_emissions_factor	T CH_4 per harvested ha per length of season	High	Medium	Low	Medium	Low	Low
N2O_emissions_factor	Not used; NA						
other_gas_emissions_factor	Not used; NA						
CO2_emissions	Not used; NA						
CH4_emissions	T CH_4	High	Medium	Medium	Medium	Medium	Low
N2O_emissions	Not used; NA						
other_gas_emissions	Not used; NA						
total_CO2e_100yrGWP	Based on CH_4 emissions	High	High	High	Medium	Medium	Low
total_CO2e_20yrGWP	Based on CH_4 emissions	High	High	High	Medium	Medium	Low
other_1	Area of single crop	Very High	Very High	Very High	Very High	Very High	Not used; NA
other_2	Area of double crop	Very High	Very High	Very High	Very High	Very High	Not used; NA
other_3	Area of triple crop	Very High	Very High	Very High	Very High	Very High	Not used; NA

Country	ISO3	Sub group	capacity	capactiy_ factor	activity	CO2_ emissions_ factor	CH4_ emissions_ factor	N2O_ emissions_ factor	other_ gas_ emissions_ factor	CO2_ emissions	CH4_ emissions	N2O_ emissions	other_gas_ emissions	total_CO2e_ 100yrGWP	total_CO2e_ 20yrGWP	other_ 1	other_ 2	other_ 3
China	CHN	1	NA	NA	High	NA	High	NA	NA	NA	High	NA	NA	High	High	Very High	Very High	Very High
Vietnam	VNM	1	NA	NA	High	NA	High	NA	NA	NA	High	NA	NA	High	High	Very High	Very High	Very High
Thailand	THA	1	NA	NA	High	NA	High	NA	NA	NA	High	NA	NA	High	High	Very High	Very High	Very High
Bangladesh	BGD	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Brazil	BRA	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Spain	ESP	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Indonesia	IDN	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
India	IND	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Iran (Islamic Republic of)	IRN	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Italy	ITA	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Japan	JPN	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Cambodia	KHM	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Korea (the Republic of)	KOR	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Lao People's Democratic Republic (the)	LAO	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Sri Lanka	LKA	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Myanmar	MMR	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Malaysia	MYS	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Nepal	NPL	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Pakistan	PAK	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Philippines (the)	PHL	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High

Country	ISO3	Sub group	capacity	capactiy_ factor	activity	CO2_ emissions_ factor	CH4_ emissions_ factor	N2O_ emissions_ factor	other_ gas_ emissions_ factor	CO2_ emissions	CH4_ emissions	N2O_ emissions	other_gas_ emissions	total_CO2e_ 100yrGWP	total_CO2e_ 20yrGWP	other_ 1	other_ 2	other_ 3
Korea (the Democratic People's Republic of)	PRK	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Taiwan (Province of China)	TWN	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
United States of America (the)	USA	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Ethiopia	ETH	5	NA	NA	Medium	NA	Low	NA	NA	NA	Medium	NA	NA	Medium	Medium	Very High	Very High	Very High
Egypt	EGY	5	NA	NA	Medium	NA	Low	NA	NA	NA	Medium	NA	NA	Medium	Medium	Very High	Very High	Very High
Turkey	TUR	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Madagascar	MDG	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Tanzania	TZA	2	NA	NA	High	NA	Medium	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Uruguay	URY	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Suriname	SUR	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Peru	PER	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Guyana	GUY	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Ecuador	ECU	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Russia	RUS	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Mexico	MEX	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Bhutan	BTN	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Bulgaria	BGR	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Burkina Faso	BFA	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Burundi	BDI	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Cameroon	CMR	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High

Country	ISO3	Sub group	capacity	capactiy_ factor	activity	CO2_ emissions_ factor	CH4_ emissions_ factor	N2O_ emissions_ factor	other_ gas_ emissions_ factor	CO2_ emissio ns	CH4_ emissions	N2O_ emissions	other_gas_ emissions	total_CO2e_ 100yrGWP	total_CO2e_ 20yrGWP	other_ 1	other_ 2	other_ 3
France	FRA	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Gambia	GMB	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Greece	GRC	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Guinea	GIN	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Guinea-Bissau	GNB	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Hungary	HUN	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Kenya	KEN	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Malawi	MWI	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Mali	MLI	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Mauritania	MRT	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Morocco	MAR	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Niger	NER	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Nigeria	NGA	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
North Macedonia	MKD	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Portugal	PRT	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Rwanda	RWA	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Senegal	SEN	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Sierra Leone	SLE	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Timor-Leste	TLS	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Uganda	UGA	3	NA	NA	High	NA	Low	NA	NA	NA	Medium	NA	NA	High	High	Very High	Very High	Very High
Backfilled Countries		6	NA	NA	Low	NA	Low	NA	NA	NA	Low	NA	NA	Low	Low	NA	NA	NA

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Geographic boundaries and names (iso3_country data attribute): The depiction and use of boundaries, geographic names and related data shown on maps and included in lists, tables, documents, and databases on Climate TRACE are generated from the Global Administrative Areas (GADM) project (Version 4.1 released on 16 July 2022) along with their corresponding ISO3 codes, and with the following adaptations:

- HKG (China, Hong Kong Special Administrative Region) and MAC (China, Macao Special Administrative Region) are reported at GADM level 0 (country/national);
- Kosovo has been assigned the ISO3 code 'XKX';
- XCA (Caspian Sea) has been removed from GADM level 0 and the area assigned to countries based on the extent of their territorial waters;
- XAD (Akrotiri and Dhekelia), XCL (Clipperton Island), XPI (Paracel Islands) and XSP (Spratly Islands) are not included in the Climate TRACE dataset;
- ZNC name changed to 'Turkish Republic of Northern Cyprus' at GADM level 0;
- The borders between India, Pakistan and China have been assigned to these countries based on GADM codes Z01 to Z09.

The above usage is not warranted to be error free and does not imply the expression of any opinion whatsoever on the part of Climate TRACE Coalition and its partners concerning the legal status of any country, area or territory or of its authorities, or concerning the delimitation of its borders.

Disclaimer: The emissions provided for this sector are our current best estimates of emissions, and we are committed to continually increasing the accuracy of the models on all levels. Please review our terms of use and the sector-specific methodology documentation before using the data. If you identify an error or would like to participate in our data validation process, please contact us.

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